

SkelGym: Cross-Paradigm Kinematic Fusion and Dynamic Augmentation for 22-Class Gym Exercise Classification

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Abstract

Automated gym exercise recognition is critical for developing intelligent virtual personal trainer systems capable of monitoring workout execution, tracking training volume, and mitigating musculoskeletal injury risks. However, traditional RGB video approaches incur prohibitive computational costs, suffer from environmental background overfitting, and raise severe privacy concerns when streaming from home environments. We evaluate on a curated corpus of 1,024 in-the-wild videos spanning 22 fine-grained resistance exercises under strict video-level partitioning with no source-video overlap across train, validation, and test sets. We present **SkelGym**, a lightweight, privacy-aware intelligent exercise recognition framework operating exclusively on 3D skeletal coordinates extracted via monocular pose estimation. SkelGym introduces four primary technical contributions: (1) a curated, multi-source 3D skeletal benchmark spanning 1,024 videos across 22 resistance exercises under strict video-level partitioning with zero source-video overlap across train, validation, and test sets; (2) a 117-dimensional *Biomechanical Compound Representation* fusing 3D hip-centered relative joint displacements (39-d) with pairwise limb elevation angles (78-d); (3) *SkelGym-Aug*, an empirically selected, task-oriented augmentation configuration specifically tailored for resistance exercise biomechanics through controlled Leave-One-Out ablation; and (4) a cross-paradigm ensemble coupling a self-attention Transformer

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with a four-stream Adaptive Graph Convolutional Network via validation-calibrated SLSQP soft voting, achieving real-time low-latency edge classification (sub-millisecond for individual backbones on CPU/CUDA). SkelGym achieves **69.73% $\pm$ 1.10% window-level accuracy** and **78.83% $\pm$ 0.66% video-level consensus accuracy** (Video Macro F1: **0.7807 $\pm$ 0.0126**, Window Macro F1: 0.6881 ± 0.0072) across three independent random seeds under fixed splits ($N = 2,743$ test windows from 233 held-out videos), with selected model comparisons evaluated via paired statistical tests (McNemar $p < 10^{-12}$, Wilcoxon $p < 0.005$). With an ultra-low classifier inference latency of merely **0.42–3.96 ms** on host CPU (0.50–4.65 ms on CUDA, 1.24–5.86 ms on MPS; where single-model backbones execute in 0.42–1.03 ms and the full five-stream ensemble executes in 3.96 ms) per 32-frame window, SkelGym enables real-time, privacy-aware exercise recognition suitable for smart mirrors, mobile devices, and embedded gym displays. All source code, extracted skeletal representations, metadata, and pre-trained checkpoints used for reproducibility are publicly available at <https://huggingface.co/Cuong2004/gym-exercise-classification>.

Keywords: Gym exercise classification, Skeleton-based action recognition, Adaptive Graph Convolutional Networks, Skeletal Transformers, Biomechanical features, Real-time inference, Edge AI

1. Introduction

Regular physical resistance training and structured exercise play an essential role in improving cardiovascular health, metabolic longevity, muscular hypertrophy, and psychological well-being [39]. With modern demanding lifestyles and remote working environments, a rapidly growing population conducts workouts in home settings rather than commercial fitness centers. However, exercising without the supervision of certified strength and conditioning coaches frequently leads to improper kinetic execution, sub-optimal muscular recruitment, and elevated risks of acute or repetitive strain injuries (such as lumbar disc herniation, rotator cuff impingement, and patellar tendinopathy) [34]. Consequently, there is an urgent demand for automated, intelligent virtual personal trainers capable of identifying exercise types in real-time, monitoring execution repetitions, and providing instantaneous biomechanical feedback [49].

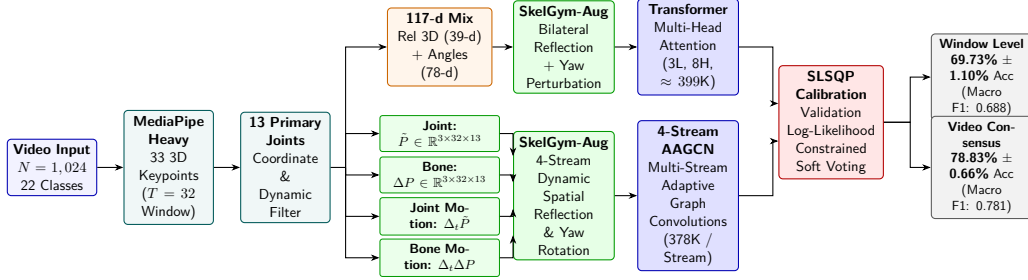


Figure 1: The proposed SkelGym architecture: an end-to-end cross-paradigm framework coupling self-attention sequence modeling with multi-stream adaptive graph convolutional networks via validation-calibrated weighted soft voting.

Early computer vision frameworks for resistance exercise and action recognition were predominantly built upon appearance-based RGB video models, such as 2D/3D CNNs [1, 2, 3] and spatiotemporal Vision Transformers (e.g., ViViT [4], TimeSformer [5], Video Swin [6]). While dense RGB representations capture visual cues such as gym equipment and barbells [7], they exhibit three critical operational drawbacks in practical deployments.

First, dense spatio-temporal convolutions incur heavy computational overhead (often exceeding hundreds of GFLOPs), which precludes real-time execution on resource-constrained mobile phones, smart displays, or edge hardware. Second, appearance-based networks readily memorize spurious background correlations—including flooring patterns, studio lighting, bench upholstery, and athletic apparel—resulting in severe performance collapse when evaluated in novel domestic workout environments. Third, cloud-centric video architectures necessitate continuously transmitting unconstrained high-definition RGB video streams from private living spaces to remote servers, introducing network bandwidth bottlenecks and user privacy vulnerabilities.

Skeleton-based human action recognition offers an elegant solution to these challenges by abstracting human bodily movement into sparse, lightweight 3D keypoint coordinates. With real-time pose estimation frameworks such as Google MediaPipe [8, 9], OpenPose [10], and AlphaPose [11], 3D skeletal tracking can be executed on commodity webcams and mobile chipsets at sub-millisecond latencies. However, translating skeletal tracking into accurate, multi-class gym exercise recognition introduces four fundamental scientific hurdles:

The first hurdle is the scarcity of standardized, fine-grained 3D skeletal gym benchmarks with strict source-video partitioning. Existing fitness

datasets are either restricted to trivial 3–5 exercises or employ random temporal window splits that permit subtle data leakage between training and testing partitions, while also lacking multi-source in-the-wild recording diversity.

The second hurdle concerns biomechanical redundancy versus kinematic expressiveness. Resistance exercises frequently involve fine-grained differences in limb trajectories and joint angles, as demonstrated when distinguishing flat, incline, and decline bench presses, or differentiating conventional deadlifts from Romanian deadlifts. While exhaustive geometric angle formulations—such as computing all three-joint triplets—might capture these subtle distinctions, they incur a combinatorial explosion of features ($\binom{13}{3} = 286$), introducing dimensional multicollinearity, gradient singularities at joint lockouts, and an acute risk of overfitting.

The third hurdle stems from the lack of task-oriented validation for skeletal data augmentation. While spatial and temporal transformations (such as rotation, mirroring, scaling, time warping, and jitter) are well-established in general action recognition, applying them naively to resistance training risks altering fixed anatomical bone lengths, violating gravitational verticality, or distorting repetition cadence and velocity profiles that define distinct exercise classes.

Finally, the fourth hurdle lies in achieving true cross-paradigm model complementarity and real-time edge execution while strictly preventing data contamination. Temporal sequence architectures (such as LSTMs [12, 13] and Transformers [14]) excel at modeling long-range phase transitions, whereas spatial-temporal graph convolutional networks (such as ST-GCN [16], 2S-AGCN [17], and CTR-GCN [20]) explicitly exploit physical musculoskeletal topology. Prior works often combine these paradigms naively, rely on heavy pre-trained backbones unsuitable for edge hardware, or tune ensemble fusion weights directly on test partitions, inducing optimistic reporting bias.

To systematically resolve these challenges, we introduce **SkelGym** (illustrated in Fig. 1), an end-to-end intelligent exercise recognition system that transforms raw monocular video into actionable exercise labels through a fully automated pipeline: *Video* $\rightarrow$ *MediaPipe 3D Skeleton* $\rightarrow$ *117-d Biomechanical Representation* $\rightarrow$ *SkelGym-Aug* $\rightarrow$ *Cross-Paradigm Classifiers* $\rightarrow$ *SLSQP Ensemble* $\rightarrow$ *Video Consensus* $\rightarrow$ *Edge Deployment*. Specifically, this work provides four primary technical contributions:

C1: Multi-Source 3D Skeletal Landmark Benchmark with Leakage-Free Partitioning... We curate and release a fine-grained 3D skeletal dataset derived from 1,024 multi-camera in-the-wild videos spanning 22 resistance training exercises across three provenance streams (public repositories, stock media, and 128 ethical author-recorded gymnasium videos). Pre-extracted into 13 load-bearing 3D landmark trajectories (1,108 action segments) in JSON, NPY, and HDF5 formats under CC BY 4.0 and MIT licenses, our benchmark establishes a strict video-level partitioning protocol (6:2:2) with zero source-video overlap, completely eliminating temporal frame-level data leakage and recording-session memorization prevalent in prior fitness benchmarks.

C2: Biomechanical Compound Representation (117-d)... We design a compact kinematic descriptor fusing 3D hip-centered relative joint coordinates (39-d) with pairwise limb elevation angles (78-d). By restricting angular components to signed elevation relative to the transverse ground plane ($\theta_{ij} = \arctan2(\Delta y_{ij}, \sqrt{\Delta x_{ij}^2 + \Delta z_{ij}^2})$), this representation provides intrinsic invariance to horizontal camera yaw rotation around the gravity vector while remaining scale-invariant. Furthermore, it eliminates gradient singularities inherent to three-joint angle formulations at limb lockouts, and achieves a 59.1% dimensional reduction over combinatorial triplets ($\binom{13}{3} = 286$) to resolve multicollinearity and mitigate MediaPipe monocular depth jitter.

C3: Empirically Selected Task-Oriented Augmentation Configuration (SkelGym-Aug)... Rather than claiming novel mathematical transformation operators, we evaluate a candidate pool of established skeletal augmentations categorized into geometric (sagittal reflection, gravitational yaw rotation, proportional scaling), temporal (time warping), and noise-based (sensor jitter) domains. Through systematic Leave-One-Out (LOO) ablation across independent random seeds, we demonstrate that while geometric operators and mild jitter supply critical viewpoint and anthropometric invariance, temporal time warping distorts repetition cadence and degrades recognition consistency. We thus isolate an optimal, task-oriented 4-operator configuration that strictly enforces musculoskeletal and gravitational constraints during training, achieving zero test-time corruption while significantly dampening optimization variance.

C4: Cross-Paradigm Fusion, Rigorous Benchmarking, and Real-Time Edge Deployment... We engineer custom lightweight backbones ($\approx 350K$ parameters each, trained entirely from scratch) spanning self-attention sequence

Transformers and four-stream Adaptive Graph Convolutional Networks (AAGCN), unified via a leakage-free SLSQP soft-voting formulation [27, 28] calibrated strictly on validation partitions. Through multi-seed evaluations and paired statistical testing (McNemar $p < 10^{-14}$, Wilcoxon $p < 0.005$), SkelGym attains **69.73% $\pm$ 1.10% window accuracy** and **78.83% $\pm$ 0.66% video consensus accuracy** (Video Macro F1: **0.7807 $\pm$ 0.0126**, Window Macro F1: 0.6881 ± 0.0072) across 22 resistance exercises. Delivering an ultra-low classifier inference latency of merely **0.42–3.96 ms** on host CPU (0.50–4.65 ms on CUDA, where single-model backbones execute in 0.42–1.03 ms and the full five-stream ensemble executes in 3.96 ms) per 32-frame window, SkelGym demonstrates practical real-time feasibility for privacy-aware edge deployment without persistent video streaming.

2. Related Work

2.1. Skeletal Action Recognition

Extracting human pose from monocular RGB video has matured rapidly with lightweight real-time estimators such as OpenPose [10], AlphaPose [11], and Google MediaPipe Pose [8, 9]. Early landmark-based action recognition utilized Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks [12, 13, 15] to model skeletal temporal dynamics. However, pure recurrent models treat skeletons as unorganized coordinate vectors, ignoring the inherent topological structure of the human musculoskeletal system.

To overcome this, Yan et al. [16] pioneered Spatial Temporal Graph Convolutional Networks (ST-GCN), constructing a spatio-temporal graph where human joints form vertices and physical bones form fixed spatial edges. Shi et al. [17] introduced Two-Stream Adaptive Graph Convolutional Networks (2S-AGCN), parameterizing the adjacency matrix with learnable and data-driven components. Subsequent advances—including multi-stream AAGCN [18], CTR-GCN [20], InfoGCN [47], HD-GCN [48], and SkateFormer [21]—progressively integrated channel attention, information-theoretic representation learning, hierarchical graph decomposition, and partition-specific transformer attention. In parallel, Duan et al. [56] introduced PoseConv3D, mapping 3D skeleton keypoints to spatiotemporal heatmap volumes to exploit 3D convolutional architectures, while Yin et al. [50] designed an efficient multi-stream depthwise separable convolutional network for skeleton action recognition. However, these modern architectures were predominantly tuned

around large-scale benchmarks like NTU RGB+D (60/120 classes) with parameter footprints of 1.3M–5.2M and high GFLOP demands. In resistance exercise tracking, peripheral joints suffer severe occlusion by barbells and machine frames. SkelGym prunes the skeleton to $V = 13$ load-bearing joints and designs custom lightweight backbones ($\approx 350\text{K}$ parameters per stream) trained from scratch, delivering real-time edge execution.

2.2. Kinematic Sequence Modeling and Skeletal Transformers

Following the dominance of self-attention in NLP and computer vision (e.g., Vaswani et al. [14], ViViT [4], TimeSformer [5], Video Swin [6]), transformers have been extended to skeletal motion (e.g., ST-TR [19]). While transformers possess global receptive fields, they are data-hungry and prone to overfitting on limited skeletal training cohorts. We demonstrate that when fed with our 117-d Biomechanical Compound Representation and regularized by SkelGym-Aug, a compact 3-layer Transformer achieves competitive performance, functioning as an ideal complement to graph convolutions.

2.3. Skeletal Data Augmentation in Action Recognition

Data augmentation is a foundational mechanism for mitigating overfitting in deep skeleton-based recognition, particularly when training cohorts are constrained [16, 18]. Candidate augmentation operators across prior literature can be broadly categorized into spatial-geometric transformations (e.g., random 3D rotations, sagittal axis flipping, coordinate scaling), temporal manipulations (e.g., temporal cropping, time warping, speed perturbation), and noise injection (e.g., additive Gaussian coordinate jitter) [23]. While these generic transformations improve general action recognition, they are frequently applied without considering domain-specific biomechanical validity: unconstrained 3D rotations alter gravitational alignment, non-uniform scaling distorts fixed biological bone lengths, and arbitrary time warping can destroy execution tempo.

In this work, we do not claim operator-level novelty for these mathematical transformations, which are established in the literature. Rather, our contribution lies in formulating a disciplined candidate pool across geometric, temporal, and noise-based domains, and conducting a rigorous empirical Leave-One-Out ablation study to discover which transformations benefit resistance training recognition and which degrade it. This analysis reveals that while bilateral reflection, gravitational yaw rotation, proportional scaling, and mild jitter are mutually synergistic, temporal time warping impairs

the model’s ability to discriminate exercise cadence and pacing. The resulting empirically selected *SkelGym-Aug* configuration provides a tailored, biomechanically grounded regularizer for gymnasium motion analysis.

2.4. Ensemble Learning for Skeleton-Based Recognition

Multi-stream fusion has emerged as a dominant paradigm in skeleton-based action recognition. Shi et al. [18] and Yin et al. [50] demonstrated that combining joint, bone, joint-motion, and bone-motion streams via late fusion substantially improves recognition accuracy by capturing complementary kinematic perspectives. However, existing multi-stream approaches typically employ either uniform averaging or manually tuned stream weights. Furthermore, when ensemble calibration involves test partition data—whether explicitly or implicitly through iterative hyperparameter sweeps—the resulting accuracy estimates suffer from optimistic reporting bias [29]. SkelGym addresses both limitations through a principled SLSQP-based constrained optimization formulation [27, 28] that calibrates stream voting weights exclusively on validation log-likelihoods, ensuring strictly held-out evaluation integrity.

2.5. Fitness and Resistance Exercise Recognition

Automated exercise tracking has attracted growing attention in sports science, rehabilitation, and ubiquitous computing. García-de-Villa et al. [49] developed an expert system for simultaneous exercise recognition and evaluation in prescribed fitness routines for virtual coaches. Strömbäck et al. [51] created MM-Fit, a multimodal benchmark unifying 3D pose, multi-view RGB-D video, and wearable inertial sensors across free-weight exercises (e.g., squats, lunges, and curls), demonstrating the utility of multi-sensor tracking. Riccio [34] implemented real-time exercise classification and repetition counting using a BiLSTM over geometric joint angles. Ding et al. [36] developed the RepCount benchmark for repetition counting in diverse fitness environments. Deyzel et al. [46] proposed one-shot skeleton-based recognition for strength and conditioning exercises on the SU-EMD dataset (5 exercises), revealing critical confusion between squats and deadlifts due to kinematic overlap. While general benchmarks (UCF101 [37], Kinetics-400 [38]) feature broad athletic activities, they lack fine-grained biomechanical resolution. Existing fitness datasets are either restricted to trivial 3–5 exercises or employ random temporal window splits that permit data leakage. In contrast, SkelGym benchmarks a demanding 22-class resistance exercise taxonomy under

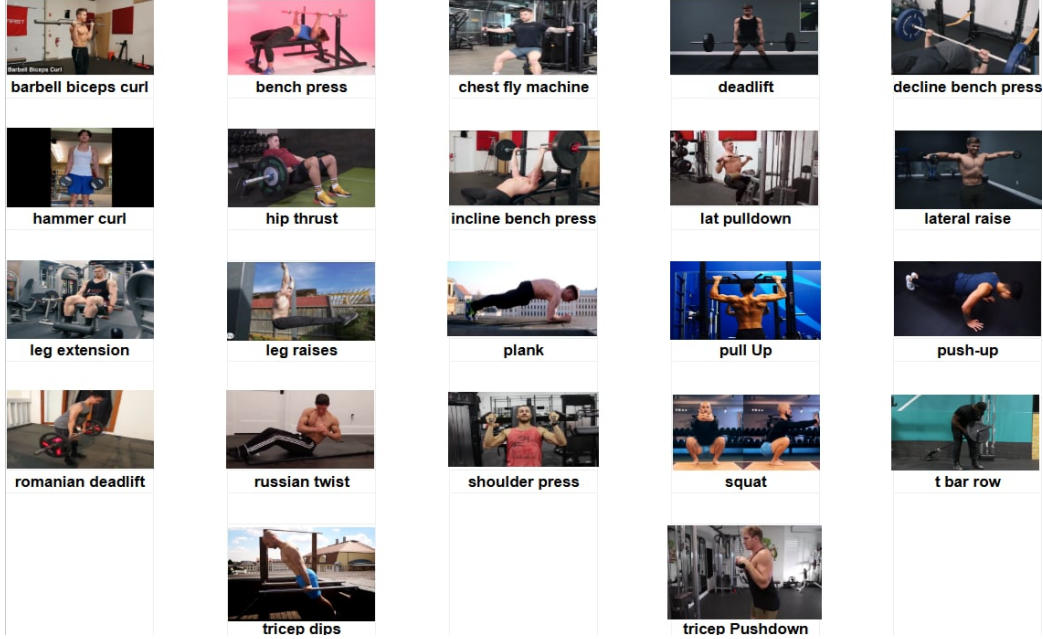


Figure 2: Taxonomy and sample distributions across the 22 resistance exercises in the benchmark dataset.

a strictly segregated video-level split protocol, advancing beyond laboratory-controlled settings to in-the-wild multi-camera gym environments.

3. Dataset and Experimental Protocol

3.1. Data Source, Provenance, and Ethical Compliance

Our experimental corpus consists of 1,024 clean, un-shuffled video recordings across 22 resistance training exercises: *barbell biceps curl*, *bench press*, *chest fly machine*, *deadlift*, *decline bench press*, *hammer curl*, *hip thrust*, *incline bench press*, *lat pulldown*, *lateral raise*, *leg extension*, *leg raises*, *plank*, *pull-up*, *push-up*, *Romanian deadlift*, *Russian twist*, *shoulder press*, *squat*, *t-bar row*, *tricep dips*, and *tricep pushdown* (Fig. 2). To ensure high diversity in environmental backgrounds, camera angles, lighting conditions, and athlete anthropometry, the corpus unifies three distinct provenance streams: (1) *public video repositories*, comprising 652 videos recovered from the open Workout/Exercise Video Dataset [35]; (2) *curated online media*, including 103 high-definition clips collected from YouTube fitness channels, 65 clips

Table 1: Key specifications, provenance breakdown, and capture characteristics of the GYMED exercise dataset.

Specification / Characteristic	Value / Distribution
Total Videos / Total File Volume	1,024 unique video recordings / 10,225.37 MB ($\approx$ 10.2 GB)
File Formats & Video Encoding	MP4 (964 videos, 94.1%), MOV (60 videos, 5.9%) / H.264
Frame Rate & Recording Duration	30 frames per second (FPS) / 20–30 s per recording
Trimmed Action Segments	1,108 clean exercise segments (average 1.09 segments per video)
Number of Distinct Resolutions	33 resolutions (1280 $\times$ 720: 385; 1920 $\times$ 1080: 315; 720 $\times$ 1280: 130; others: 194)
Exercise Taxonomy Coverage	22 fine-grained resistance exercise classes
Video Partitioning Ratio (6:2:2)	Train: 580 (56.6%), Validation: 208 (20.3%), Test: 236 (23.1%)
Action Segment Partitions	Train: 639 (57.7%), Validation: 210 (19.0%), Test: 259 (23.3%)
Temporal Windows ($T = 32$ frames)	Train: 13,136 ($S = 16$), Validation: 2,075 ($S = 32$), Test: 2,743 ($S = 32$)
Multi-Source Data Provenance	Abdillah (2023) [35]: 652; YouTube: 103; Pexels: 65; Freepik: 76; Authors: 128
Subjects & Ethical Compliance	Informed consent under Declaration of Helsinki for recorded clips; diverse online athletes
Camera Dynamics & Perspectives	Static tripod and dynamic moving cameras; frontal, sagittal, oblique angles
Recording Environments	Commercial gymnasiums, domestic home workout spaces, outdoor fitness zones

from Pexels, and 76 clips from Freepik acquired under open research and stock licenses; and (3) *author-recorded footage*, consisting of 128 supplementary videos captured by the authors in a gymnasium across all 22 exercise classes with two athletic models, employing both static tripod viewpoints and dynamic moving-camera angles to capture multi-perspective kinetic execution.

Subject Statistics and Ethical Compliance.. To provide both high-precision kinematic reference baselines and expansive real-world diversity, the dataset incorporates two complementary subject cohorts. For the 128 self-recorded gymnasium videos, volunteer participants executed all 22 resistance movements under controlled multi-angle camera setups, providing kinematically disciplined execution benchmarks under formal written informed consent following the Declaration of Helsinki. For the remaining 896 web-sourced videos, recordings capture hundreds of diverse athletes spanning unconstrained body builds, athletic apparel, and training experience across international fitness facilities, home gyms, and outdoor parks. Crucially, the dataset does not annotate explicit participant demographic or gender identifiers, ensuring that model training and evaluation remain strictly posture-driven and invariant to demographic bias. Furthermore, the video-level partition ensures that 100% of video clips and derived segments from any given recording session reside entirely within a single split, guaranteeing zero source-video overlap and eliminating temporal leakage across train, validation, and test sets. We note that while web-sourced video repositories lack explicit biometric participant identifiers, isolating all segmented windows strictly by source video guarantees that models cannot memorize repetitive background context, camera

calibration artifacts, or session-specific clothing.

Source Statistics and Video Formats.. As summarized in Table 1, the 1,024 video corpus totals 10,225.37 MB (≈ 10.2 GB) in storage, distributed between MP4 (964 videos, 94.1%) and MOV (60 videos, 5.9%) containers using H.264 compression. All videos operate at 30 frames per second. Video resolution was verified via `ffprobe`, encompassing 33 distinct resolutions dominated by standard high-definition formats: 1280×720 (385 videos, 37.6%), 1920×1080 (315 videos, 30.8%), and portrait mobile video 720×1280 (130 videos, 12.7%), with 35 videos at 854×480 , 28 at 640×360 , 23 at 3840×2160 (4K), and 108 videos distributed across 27 minor resolutions (≤ 20 videos each).

Ethical Standards and Informed Consent.. For the 128 self-recorded videos, written informed consent was formally obtained from all participating human subjects prior to filming. The data recording protocol was conducted in strict adherence to institutional ethical standards and the Declaration of Helsinki. For data harvested from public online platforms, collection adhered strictly to platform Terms of Service and redistribution guidelines, gathering only publicly disseminated content without collecting private personal identifying information.

Data Licensing and Artifact Availability.. To respect the intellectual property and terms of external video hosting platforms (YouTube, Pexels, and Freepik), original raw third-party video recordings are not redistributed. Their original source identifiers, attribution information, and manual segmentation boundaries are retained within the metadata manifests. Instead, to guarantee 100% scientific reproducibility, all derived scientific artifacts—including extracted 3D skeletal landmark coordinate tensors, manual action segmentation boundaries, preprocessed kinematic features, and metadata manifests—are publicly released under the Creative Commons Attribution 4.0 International license (CC BY 4.0). All neural network architectures, training routines, and pre-trained checkpoints are openly provided under the permissive MIT License at <https://huggingface.co/Cuong2004/gym-exercise-classification>.

Video Inclusion and Exclusion Criteria.. To ensure high-quality skeletal extraction and biomechanically meaningful training samples, we applied the following curation protocol. Videos were *included* if they (i) depicted a clearly identifiable single athlete performing one of the 22 target exercises, (ii) were

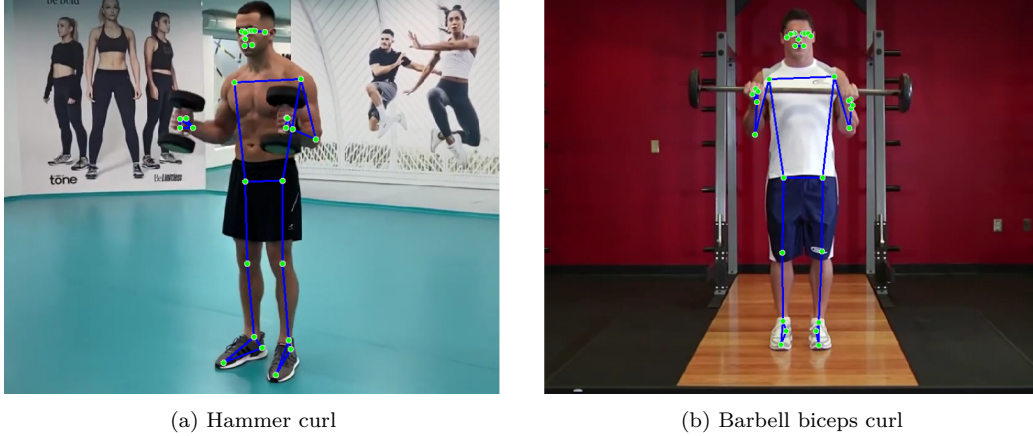


Figure 3: Full-body 3D pose landmark detection results across representative exercise video frames using Google MediaPipe Pose: (a) hammer curl and (b) barbell biceps curl.

recorded at a frame rate of ≥ 25 FPS, (iii) contained at least one complete repetition cycle with visible concentric and eccentric phases, and (iv) maintained sufficient camera framing such that hip and shoulder joints remained visible for $\geq 80\%$ of the active exercise duration. Conversely, videos were *excluded* if they exhibited (a) near-complete occlusion of both hip or both shoulder joints throughout the exercise segment (e.g., extreme close-up shots of isolated limbs), (b) a native frame rate below 25 FPS (resulting in temporal aliasing during 30 FPS resampling), (c) an active exercise duration shorter than one full repetition cycle (< 32 frames after trimming), or (d) severe motion blur or compression artifacts rendering MediaPipe Pose landmark detection unreliable (landmark visibility confidence < 0.3 for $> 50\%$ of frames). This protocol yielded the final curated corpus of 1,024 videos described above.

3.2. Action Trimming, Video-Level Partitioning, and Class Distribution

Raw workout recordings frequently contain non-exercise transitional intervals, such as picking up barbells, adjusting bench pins, chalking hands, and resting. To eliminate noise, all 1,024 videos were manually verified and trimmed into action-only segments via timestamp boundary files (‘.txt’), yielding 1,108 clean segments total (1.09 segments per video on average).

Strict Video-Level Partitioning Protocol (No Source-Video Overlap Across Splits).. To ensure rigorous evaluation without data leakage, we enforce a

strict video-level partition in a 6:2:2 ratio. Our data processing follows a strictly segregated 4-step pipeline: (1) *Video-Level Isolation*: Raw video recordings are partitioned strictly by source video file into Train ($N = 580$), Validation ($N = 208$), and Test ($N = 236$) sets, ensuring no video file or recording session spans across partitions. (2) *Independent Landmark Extraction*: Monocular 3D skeletal landmarks are extracted independently per video file using Google MediaPipe Pose without inter-video context. (3) *Within-Video Sliding Window Segmentation*: Temporal windows ($T = 32$ frames) are extracted strictly within individual trimmed action boundaries: $S = 16$ frames (50% overlap) for Train (13,136 windows) to provide dense temporal coverage, and $S = 32$ frames (non-overlapping) for Validation (2,075 windows) and Test (2,743 windows across 233 videos with valid segments ≥ 32 frames). No sliding window is permitted to cross video or segment boundaries. (4) *Strict Training-Partition Feature Standardization*: Joint spatial coordinates are centered relative to the mid-hip origin ($p_{\text{hip_mid}} = \frac{1}{2}(p_{\text{left_hip}} + p_{\text{right_hip}})$) per frame, establishing translation invariance. To scale features uniformly without data leakage, global feature-wise z -score normalization statistics ($\mu_{\text{train}}, \sigma_{\text{train}}$) are computed strictly on the training partition and frozen, then applied to normalize training, validation, and test representations (with on-the-fly physical augmentations strictly preceding standardization). This protocol ensures zero source-video overlap, zero temporal window leakage, and zero test-information contamination in feature normalization. Because web-sourced exercise recordings lack explicit biometric subject identifiers, this strict video-level partition guarantees that all sub-clips, temporal windows, and repetitions from any single recording session reside exclusively within a single partition (Train, Validation, or Test), eliminating both temporal sequence leakage and intra-session environment memorization.

Class Imbalance and Metric Selection.. Table 2 details the distribution of videos and action segments across all 22 classes. As shown, natural gym workout data exhibits significant class imbalance: highly popular compound movements such as *barbell biceps curl* ($N = 84$ videos) and *bench press* ($N = 84$ videos) possess large sample representations, whereas specialized exercises such as *plank* ($N = 9$ videos total: 5 train, 2 val, 2 test) are relatively scarce. To prevent models from artificially inflating overall accuracy by predicting majority classes, we adopt **Macro-averaged F1 score** alongside Accuracy as the primary evaluation benchmark.

Table 2: Video and action segment distributions across the 22 resistance training exercises by partition split.

Exercise Class	Videos				Action Segments			
	Train	Val	Test	Total	Train	Val	Test	Total
Barbell biceps curl	52	18	14	84	53	18	14	85
Bench press	51	18	15	84	51	18	15	84
Chest fly machine	28	9	8	45	29	9	8	46
Deadlift	30	10	10	50	30	10	10	50
Decline bench press	18	5	9	32	29	5	10	44
Hammer curl	24	8	14	46	28	8	14	50
Hip thrust	14	6	9	29	19	6	15	40
Incline bench press	27	10	9	46	27	10	9	46
Lat pulldown	42	15	14	71	42	15	15	72
Lateral raise	24	8	15	47	24	8	15	47
Leg extension	22	9	13	44	27	9	13	49
Leg raises	15	6	11	32	15	6	11	32
Plank	5	2	2	9	5	2	2	9
Pull-up	31	11	10	52	31	11	10	52
Push-up	40	14	12	66	43	14	14	71
Romanian deadlift	12	6	6	24	13	6	8	27
Russian twist	15	6	6	27	19	6	6	31
Shoulder press	27	10	13	50	28	10	14	52
Squat	24	10	15	49	27	11	17	55
T-bar row	15	4	10	29	17	4	10	31
Tricep pushdown	39	14	12	65	39	14	12	65
Tricep dips	25	9	9	43	43	10	25	78
Total	580	208	236	1,024	639	210	259	1,108

Data and Code Availability.. To promote open science and persistent reproducibility, all extracted 3D pose landmark sequences (in JSON, NPY, and HDF5 tensor formats), action segmentations (`.txt`), partition manifests, and Python code are publicly hosted on Hugging Face Hub at <https://huggingface.co/Cuong2004/gym-exercise-classification>. Distributing pre-extracted coordinates circumvents vulnerability to third-party URL link rot common in video scraping, protects athlete privacy, and ensures permanent deterministic reproducibility. Raw third-party internet video recordings are withheld in adherence to platform Terms of Service, while the 128 author-recorded gymnasium clips and complete skeletal tensors are freely accessible

for non-commercial academic research (see Appendix [Appendix B](#)). **Licensing:** The extracted 3D landmark coordinate tensors (NPY, JSON, HDF5), action segmentation boundaries, and partition manifests are released under the **Creative Commons Attribution 4.0 International (CC BY 4.0)** license. All source code, training scripts, evaluation pipelines, and configuration files are released under the **MIT License**.

3.3. Landmark Extraction and Anatomical Joint Selection

Every video frame is processed through Google MediaPipe Pose Heavy (complexity level 2) [8, 9], extracting 33 full-body 3D landmarks $(x_i, y_i, z_i, \text{vis}_i)$ normalized relative to bounding box and depth dimensions, as illustrated on representative workout frames in Fig. 3. Because facial landmarks and fine-grained finger points contribute negligible kinetic leverage to compound resistance exercises while introducing tracking jitter, we filter the skeleton to $V = 13$ major anatomical joints (detailed in Table A.16): nose (cranial orientation tracking), left/right shoulders, left/right elbows, left/right wrists, left/right hips (pelvic centroid / origin anchor), left/right knees, and left/right ankles.

4. Proposed Methodology

The overall architectural workflow of **SkelGym** is illustrated in Fig. 1, while Fig. 4 details the underlying biomechanical feature extraction and kinematic tensor decomposition. Below, we formally derive each mathematical component of our framework.

4.1. Problem Formulation and Mathematical Notation

Let $\mathcal{V} = \{I_t\}_{t=1}^{N_V}$ represent a trimmed resistance workout video of N_V consecutive RGB frames, where $I_t \in \mathbb{R}^{H \times W \times 3}$. A monocular pose estimation operator $\Phi_{\text{pose}} : I_t \mapsto \mathcal{P}_t$ extracts $V_0 = 33$ three-dimensional body landmarks $\mathcal{P}_t = \{(x_{t,v}, y_{t,v}, z_{t,v}, \nu_{t,v})\}_{v=1}^{V_0}$, where $(x, y, z) \in [-1, 1]^3$ represent normalized spatial coordinates and $\nu \in [0, 1]$ is a visibility confidence indicator.

To eliminate tracking jitter and high-frequency noise from distal extremities (such as facial landmarks and toes) that contribute negligible physical leverage to compound resistance movements, we apply an anatomical pruning operator $\Pi_{13} : \mathbb{R}^{V_0 \times 4} \mapsto \mathbb{R}^{V \times 3}$ retaining $V = 13$ primary structural joints

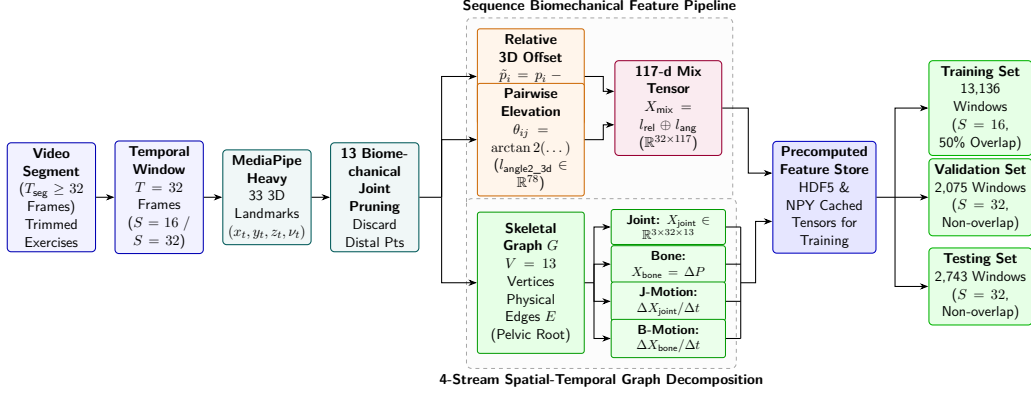


Figure 4: The proposed Biomechanical Feature Extraction and Kinematic Tensor Decomposition pipeline: illustrating the two-branch extraction of the 117-d sequence mix and the 4-stream spatial-temporal graph tensors.

(see Appendix [Appendix A](#)): the nose (cranial orientation tracking), bilateral shoulders, elbows, wrists, hips (pelvic centroid origin anchor), knees, and ankles.

For each action segment, temporal sampling generates fixed-length windows $w_k \in \mathbb{R}^{T \times V \times 3}$ spanning $T = 32$ frames (≈ 1.07 seconds at 30 FPS). Let $\mathcal{Y} = \{1, \dots, C\}$ denote the label space corresponding to the $C = 22$ gym exercise categories. The primary learning objective is to parameterize a predictive distribution $P_{\Theta}(y = c \mid w)$ that maximizes the likelihood of the true exercise class $y \in \mathcal{Y}$, and subsequently infer a consensus video-level decision $\hat{y}_{\text{vid}}$ across all K_v constituent windows of video $\mathcal{V}$.

4.2. Biomechanical Compound Representation (117-d Kinematic Mix)

Directly feeding absolute skeletal coordinates $p_{t,v} \in \mathbb{R}^3$ into neural classifiers forces models to memorize camera distance, subject translation, and framing perspective. Conversely, computing full 3-joint triplet interior angles $\angle(v_i, v_j, v_k)$ across all combinations $\binom{13}{3} = 286$ induces severe collinearity, numerical instability, and parameter redundancy. To resolve this trade-off, we formulate the **Biomechanical Compound Representation** ($l_{\text{mix}} \in \mathbb{R}^{117}$) by coupling spatial translation-invariant displacements with pairwise directional elevation angles in spherical projection:

4.2.1. Relative 3D Spatial Displacements ($\mathbf{l}_{rel_3d}$):

We establish an intrinsic bodily coordinate frame by translating all key-points relative to the pelvic midpoint between the left and right hip joints (indices 7 and 8), representing the anatomical center of mass of the human body:

$$p_{t,hip_mid} = \frac{1}{2} (p_{t,left_hip} + p_{t,right_hip}) \in \mathbb{R}^3, \quad (1)$$

$$\tilde{p}_{t,v} = p_{t,v} - p_{t,hip_mid} \in \mathbb{R}^3, \quad \forall v \in \{1, \dots, 13\}. \quad (2)$$

Concatenating $\tilde{p}_{t,v}$ across all $V = 13$ joints (including the cranial nose landmark) yields a 39-dimensional translation-invariant spatial vector:

$$l_{rel_3d}(t) = [\tilde{p}_{t,1}^T, \tilde{p}_{t,2}^T, \dots, \tilde{p}_{t,13}^T]^T \in \mathbb{R}^{39}. \quad (3)$$

4.2.2. Pairwise Directional Elevation Angles ($\mathbf{l}_{angle2_3d}$):

For every unordered pair of distinct anatomical joints (i, j) with $1 \leq i < j \leq 13$ ($\binom{13}{2} = 78$ unique limb chords), we compute the directed Euclidean displacement vector:

$$\Delta p_{ij}(t) = \tilde{p}_{t,j} - \tilde{p}_{t,i} = (\Delta x_{ij}(t), \Delta y_{ij}(t), \Delta z_{ij}(t)) \in \mathbb{R}^3. \quad (4)$$

From Δp_{ij} , we compute the signed elevation angle $\theta_{ij}(t)$ relative to the horizontal transverse ground plane ($X - Z$):

$$\theta_{ij}(t) = \arctan 2 \left(\Delta y_{ij}(t), \sqrt{\Delta x_{ij}(t)^2 + \Delta z_{ij}(t)^2} \right) \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]. \quad (5)$$

The elevation angle $\theta_{ij}(t)$ quantifies the vertical inclination of limb segment (i, j) relative to the gravity axis Y . We deliberately exclude horizontal azimuth angles ($\phi_{ij} = \arctan 2(\Delta z_{ij}, \Delta x_{ij})$) from the feature descriptor: because the horizontal radius $r_{ij} = \sqrt{\Delta x_{ij}^2 + \Delta z_{ij}^2}$ is radially symmetric under rotations about the gravity axis, elevation angles are intrinsically invariant to camera yaw rotation and subject heading direction, whereas azimuth angles would introduce viewpoint-dependent noise. Concatenating $\theta_{ij}(t)$ across all 78 unique limb pairs produces the 78-dimensional angular vector:

$$l_{angle2_3d}(t) = [\theta_{1,2}(t), \theta_{1,3}(t), \dots, \theta_{12,13}(t)]^T \in \mathbb{R}^{78}. \quad (6)$$

Biomechanical Advantages and Mathematical Derivation of Pairwise Elevation Angles. The proposed 78-dimensional pairwise elevation formulation confers four fundamental advantages over conventional 3-joint triplet interior angles $\psi = \angle(v_i, v_j, v_k)$:

1. **Analytical Attenuation of Tracking Noise Propagation:** Monocular pose estimation is inherently susceptible to localized joint tracking noise and transient occlusions. Let p_{err} represent the independent probability that a joint coordinate estimate is noisy or partially occluded. A conventional interior joint angle $\psi = \angle(v_i, v_j, v_k)$ depends on three simultaneous joint positions; the probability that at least one joint in the triplet is corrupted is:

$$P_{\text{err}}^{(3)} = 1 - (1 - p_{\text{err}})^3 = 3p_{\text{err}} - 3p_{\text{err}}^2 + p_{\text{err}}^3 \approx 3p_{\text{err}}. \quad (7)$$

In contrast, our pairwise elevation angle θ_{ij} depends on only two landmarks (i, j) , yielding an error exposure probability of:

$$P_{\text{err}}^{(2)} = 1 - (1 - p_{\text{err}})^2 = 2p_{\text{err}} - p_{\text{err}}^2 \approx 2p_{\text{err}}. \quad (8)$$

Pairwise parameterization thus yields an analytical **33.3% reduction in error susceptibility** $((3p_{\text{err}} - 2p_{\text{err}})/3p_{\text{err}} \approx 1/3)$, providing significantly greater resilience against monocular landmark jitter and optical occlusion.

2. **Elimination of Numerical Gradient Singularities at Joint Lockout:** Conventional interior joint angles are computed via vector inner products:

$$\psi = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}\right), \quad \text{with derivative} \quad \frac{d\psi}{dx} = -\frac{1}{\sqrt{1-x^2}}, \quad (9)$$

where $x = (\mathbf{u} \cdot \mathbf{v})/(\|\mathbf{u}\| \|\mathbf{v}\|) \in [-1, 1]$. In resistance exercises, full joint extension and mechanical lockout (e.g., knee lockout at the apex of a squat, elbow extension in bench press or triceps pushdown) frequently drive limb vectors into near-collinear alignment ($x \rightarrow \pm 1$). As $x \rightarrow \pm 1$, the derivative diverges ($d\psi/dx \rightarrow -\infty$), causing exploding gradients, loss spikes, and numerical instability during back-propagation. Conversely, our pairwise elevation formulation computes

$\theta_{ij} = \arctan 2(\Delta y_{ij}, r_{ij})$ where $r_{ij} = \sqrt{\Delta x_{ij}^2 + \Delta z_{ij}^2}$. Its partial derivatives:

$$\frac{\partial \theta_{ij}}{\partial \Delta y_{ij}} = \frac{r_{ij}}{\|\Delta p_{ij}\|^2}, \quad \frac{\partial \theta_{ij}}{\partial r_{ij}} = -\frac{\Delta y_{ij}}{\|\Delta p_{ij}\|^2}, \quad (10)$$

are continuous, smooth, and bounded for all non-coincident anatomical pairs ($\|\Delta p_{ij}\| > 0$). This guarantees numerical stability throughout all concentric lockout and eccentric stretching phases.

3. **Kinematic Encoding via Virtual Inter-Limb Chords:** Unlike rigid anatomical tree models that only measure physically connected bones, the 78 unordered pairs incorporate non-adjacent *virtual inter-limb chords*:

- The bilateral wrist-to-wrist chord (5, 6) directly tracks grip distance and barbell inclination relative to gravity, instantly distinguishing rigid bilateral barbell exercises (fixed grip width, synchronized vertical translation) from unilateral or independent dumbbell movements.
- Axial cross-body chords (such as shoulder-to-ankle (1, 11) and hip-to-ankle (7, 11)) measure global trunk lean and whole-body inclination relative to the ground, cleanly separating standing vertical movements (e.g., overhead shoulder press, lateral raise) from prone horizontal exercises (e.g., push-up, plank, bench press).

4. **Dimensional Compactness and Rank Preservation:** Computing full 3-joint triplets across 13 joints yields $\binom{13}{3} = 286$ features. Because triplets sharing common bones induce severe multicollinearity and rank deficiency, they overwhelm compact classifiers ($\approx 350K$ parameters). The 78 pairwise elevation angles provide a complete directional descriptor in the gravitational field with a **72.7% reduction in angular dimensionality** relative to all triplets ($286 \rightarrow 78$).

4.2.3. Compound Fusion Tensor ($\mathbf{X}_{mix}$):

The combined biomechanical descriptor at frame t is formed via direct vector concatenation:

$$l_{mix}(t) = l_{rel_3d}(t) \oplus l_{angle2_3d}(t) \in \mathbb{R}^{39+78} = \mathbb{R}^{117}. \quad (11)$$

Over the sliding window of $T = 32$ frames, the sequence input tensor is structured as:

$$\mathbf{X}_{mix} = [l_{mix}(1), l_{mix}(2), \dots, l_{mix}(32)]^T \in \mathbb{R}^{32 \times 117}. \quad (12)$$

Compared to full 3-point angle triplets (286-d) or concatenating 3D relative coordinates with triplets (325-d), our 117-d formulation reduces feature dimensionality by 64.0%, simultaneously eliminating the rank deficiency and gradient singularities of collinear joint angles while retaining both metric Euclidean distance and scale-invariant directional orientation.

4.3. Task-Oriented Skeletal Augmentation Configuration (SkelGym-Aug)

Training deep sequence and graph neural networks on skeletal trajectories introduces significant overfitting risks due to athlete-specific body dimensions, individual execution styles, and camera viewing orientations. While data augmentation is standard practice in human action recognition, applying unconstrained transformations to resistance exercises risks violating anatomical joint limits, altering gravitational loading directions, or distorting repetition tempos. To address this, we formulate a disciplined candidate pool of five established skeletal augmentation operators, categorized into geometric, temporal, and noise-based transformations, and determine their final composition through empirical Leave-One-Out ablation.

4.3.1. Candidate Transformation Pool

The candidate augmentation pool spans five operators across three physical domains:

1. Geometric Transformations..

- **Sagittal Bilateral Reflection ($\mathbb{Z}_2$ Symmetry):** Human anatomy exhibits reflective bilateral symmetry across the sagittal plane. Let $\sigma \in \mathbb{Z}_2$ denote the sagittal reflection operator acting on spatial coordinates: $\sigma(x, y, z) = (-x, y, z)$. Bilateral anatomical joints are permuted via an exact permutation involution $\pi \in \mathfrak{S}_{13}$ that swaps contralateral counterparts while preserving axial medial keypoints:

$$\pi = (0)(1\ 2)(3\ 4)(5\ 6)(7\ 8)(9\ 10)(11\ 12), \quad (13)$$

corresponding to nose ($0 \leftrightarrow 0$), shoulders ($1 \leftrightarrow 2$), elbows ($3 \leftrightarrow 4$), wrists ($5 \leftrightarrow 6$), hips ($7 \leftrightarrow 8$), knees ($9 \leftrightarrow 10$), and ankles ($11 \leftrightarrow 12$). The reflected spatial keypoints are defined as $\tilde{p}'_{t,v} = \sigma(\tilde{p}_{t,\pi(v)}) = (-\tilde{x}_{t,\pi(v)}, \tilde{y}_{t,\pi(v)}, \tilde{z}_{t,\pi(v)})$. Reflected limb elevation angles are evaluated in physical coordinate space prior to standardization, mapping through

the pair permutation involution with appropriate chord orientation parity: $\theta'_k = s_k \theta_{\pi_{\text{pair}}(k)}$, applied with activation probability $p_{\text{flip}} = 0.5$ to simulate bilateral camera viewing.

- **Gravitational Yaw Rotation (SO(2) Invariance):** To simulate horizontal camera viewpoints around the athlete without altering the gravitational vertical axis y , coordinates are rotated about the vertical axis by continuous random angle $\alpha \sim \mathcal{U}(-15^\circ, +15^\circ)$ using the planar rotation matrix $R_y(\alpha)$:

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}. \quad (14)$$

Because $R_y(\alpha)$ acts strictly within the transverse plane, vertical displacement remains invariant ($\Delta y' = \Delta y$) and horizontal Euclidean radius is identically preserved:

$$\Delta x'^2 + \Delta z'^2 = (\Delta x \cos \alpha + \Delta z \sin \alpha)^2 + (-\Delta x \sin \alpha + \Delta z \cos \alpha)^2 = \Delta x^2 + \Delta z^2. \quad (15)$$

Consequently, elevation angles satisfy $\theta' = \theta$, preserving biological limb inclination relative to gravity while teaching the network viewpoint invariance.

- **Spatial Proportional Scaling:** Uniform isotropic scaling with factor $s \sim \mathcal{U}(0.90, 1.10)$ is applied to $\tilde{p}$ to emulate varying athlete statures while strictly preserving anthropometric bone length ratios.

2. Temporal Transformations..

- **Temporal Time Warping:** Candidate execution speed perturbation is applied via linear temporal resampling of the 32-frame sequence with a warping coefficient $s_t \sim \mathcal{U}(0.85, 1.15)$, followed by uniform interpolation back to $T = 32$ frames to simulate varying repetition cadence.

3. Noise-Based Perturbations..

- **Gaussian Sensor Jitter:** Zero-mean Gaussian coordinate jitter $\epsilon \sim \mathcal{N}(0, \sigma^2 I_3)$ with $\sigma = 0.008$ is injected into skeletal coordinates prior to z -score normalization (with derived pairwise elevation angles and directional vectors dynamically recomputed for compound representations) to emulate high-frequency monocular tracking noise and video compression artifacts.

4.3.2. Pre-specified Design and Final SkelGym-Aug Configuration

Rather than combining candidate transformations ad-hoc, the **SkelGym-Aug configuration** was pre-specified on domain-first principles to target the primary sources of physical variance in monocular gym tracking—viewpoint variations, bilateral execution symmetry, anthropometric stature, and high-frequency keypoint jitter—while deliberately avoiding synthetic cadence perturbations (time warping) that alter repetition tempo and concentric-to-eccentric phase ratios. As systematically analyzed through Leave-One-Out (LOO) and single-component ablations (Section 5.3), this 4-operator protocol (*Bilateral Mirroring + Gravitational Yaw + Proportional Scaling + Sensor Jitter*) delivers strong training stability and out-of-distribution transfer across diverse neural backbones.

Zero Test-Time Data Corruption Protocol. All transformations within the SkelGym-Aug pipeline are executed *strictly on-the-fly during training forward passes* as stochastic data regularization. Validation and held-out test splits remain completely pristine and unaugmented throughout model selection, hyperparameter tuning, and final evaluation, ensuring zero test-time corruption and guaranteeing that reported metrics reflect authentic generalization to unmanipulated video recordings.

4.4. Self-Attention Sequence Modeling (Transformer Mix)

The augmented sequence tensor $X_{\text{mix}} \in \mathbb{R}^{32 \times 117}$ is processed by a temporal Transformer encoder [14] designed to capture long-range phase transitions and inter-frame kinetic dependencies across the workout duration. The encoder architecture consists of four sequential processing stages: linear projection, temporal positional encoding, stacked multi-head self-attention with feed-forward sub-layers, and global temporal pooling.

Linear Embedding and Positional Encoding. At each discrete time step $t \in \{1, \dots, 32\}$, the 117-dimensional compound feature vector is mapped into a latent embedding space of dimension $d_{\text{model}} = 128$:

$$E_0 = X_{\text{mix}} W_{\text{proj}} + b_{\text{proj}} \in \mathbb{R}^{32 \times 128}, \quad (16)$$

where $W_{\text{proj}} \in \mathbb{R}^{117 \times 128}$ and $b_{\text{proj}} \in \mathbb{R}^{128}$ are trainable parameters. Because the core self-attention operator is inherently permutation-invariant, sequence

chronology is explicitly preserved by superimposing learnable temporal positional embeddings $E_{\text{pos}} \in \mathbb{R}^{32 \times 128}$ initialized via a normal distribution ($\sigma = 0.02$):

$$Z^{(0)} = E_0 + E_{\text{pos}} \in \mathbb{R}^{32 \times 128}. \quad (17)$$

This formulation enables the model to adaptively capture phase-dependent kinematic progressions throughout repetition cycles.

Multi-Head Self-Attention.. The embedded sequence is propagated through $L = 3$ identical Transformer encoder layers. Each layer incorporates a multi-head self-attention (MHSA) module configured with $h = 8$ parallel attention heads ($d_k = d_v = 16$). For head $j \in \{1, \dots, h\}$, queries, keys, and values are generated via linear projections $Q_j = ZW_{Q,j}$, $K_j = ZW_{K,j}$, and $V_j = ZW_{V,j}$:

$$\text{head}_j = \text{Softmax} \left(\frac{Q_j K_j^T}{\sqrt{d_k}} \right) V_j, \quad (18)$$

$$\text{MHSA}(Z) = [\text{head}_1 \oplus \dots \oplus \text{head}_h] W_O, \quad (19)$$

where $W_O \in \mathbb{R}^{128 \times 128}$ aggregates the concatenated multi-head outputs into the model embedding dimension.

Residual Normalization and Feed-Forward Blocks.. To ensure stable gradient propagation and prevent representation collapse, each sub-layer employs residual skip connections coupled with pre-layer normalization:

$$Z'^{(l)} = Z^{(l-1)} + \text{MHSA} \left(\text{LayerNorm}(Z^{(l-1)}) \right), \quad (20)$$

$$Z^{(l)} = Z'^{(l)} + \text{FFN} \left(\text{LayerNorm}(Z'^{(l)}) \right). \quad (21)$$

Here, the position-wise feed-forward network is parameterized as $\text{FFN}(u) = \text{GELU}(uW_1 + b_1)W_2 + b_2$ using Gaussian Error Linear Units [25], with an expansion hidden dimension of $d_{\text{ff}} = 192$ and dropout rate of 0.2.

Global Pooling and Classification Head.. Following the final layer L , the frame representations are temporally summarized via uniform average pooling, $\hat{z} = \frac{1}{T} \sum_{t=1}^T Z_t^{(L)} \in \mathbb{R}^{128}$, and projected to normalized class posterior probabilities:

$$\hat{P}_{\text{trans}} = \text{Softmax}(\hat{z}W_{\text{cls}} + b_{\text{cls}}) \in \Delta^{21}. \quad (22)$$

4.5. Multi-Stream Spatial-Temporal Adaptive Graph Convolution (4-Stream AAGCN)

While Transformers model global sequence dynamics, physical musculoskeletal connectivity is naturally represented as a spatio-temporal graph $G = (V, E)$ [16], where vertices $V = \{1, \dots, 13\}$ denote joints and edges E denote physical bones. To capture rich kinematic perspectives, we formulate four parallel input streams following multi-stream principles [17, 18]:

$$X_{\text{joint}}(c, t, v) = \tilde{p}_{t,v}^{(c)} \in \mathbb{R}^{3 \times T \times V}, \quad (23)$$

$$X_{\text{bone}}(c, t, v) = \tilde{p}_{t,v_{\text{target}}}^{(c)} - \tilde{p}_{t,v_{\text{source}}}^{(c)} \in \mathbb{R}^{3 \times T \times V}, \quad (24)$$

$$X_{\text{j-motion}}(c, t, v) = X_{\text{joint}}(c, t+1, v) - X_{\text{joint}}(c, t, v) \in \mathbb{R}^{3 \times (T-1) \times V}, \quad (25)$$

$$X_{\text{b-motion}}(c, t, v) = X_{\text{bone}}(c, t+1, v) - X_{\text{bone}}(c, t, v) \in \mathbb{R}^{3 \times (T-1) \times V}, \quad (26)$$

where $(v_{\text{source}}, v_{\text{target}})$ represents directed bone edges originating from the pelvic center.

Each stream is independently processed by a lightweight calibrated Adaptive Graph Convolutional Network (AAGCN) [18]. For a given skeletal representation $X \in \mathbb{R}^{B \times C \times T \times V}$, the spatial graph convolution is formulated as:

$$H_{\text{spatial}} = \sigma(\text{BN}(((A_{\text{topo}} \odot M) + B + C(X))XW)), \quad (27)$$

where spatial connectivity is modulated by three complementary components:

1. $A_{\text{topo}} \in \mathbb{R}^{V \times V}$ is the normalized static anatomical adjacency matrix with self-loops ($\tilde{A} = D^{-1/2}(A + I)D^{-1/2}$) representing physical musculoskeletal bone connections, modulated element-wise by a learnable edge importance mask $M \in \mathbb{R}^{V \times V}$.
2. $B \in \mathbb{R}^{V \times V}$ is a global learnable continuous dependency matrix initialized with small Gaussian weights ($\sigma = 10^{-4}$), enabling the network to discover non-local joint correlations (e.g., wrist-ankle synchronization during deadlifts).
3. $C(X) \in \mathbb{R}^{V \times V}$ is an instance-specific, dynamic graph attention matrix computed via normalized dot-product attention over temporally pooled joint representations:

$$C(X) = \text{Softmax}\left(\frac{\theta(X)^T \phi(X)}{\sqrt{d_e}}\right), \quad (28)$$

where $\theta(X), \phi(X) \in \mathbb{R}^{V \times d_e}$ project pooled joint features into an embedding space of dimension $d_e = 16$.

Spatial graph convolution is paired with a temporal convolution (9×1 temporal kernel, padding 4), batch normalization, dropout (0.2), and residual shortcut connections with GELU activation. The backbone is structured into a calibrated 3-stage hierarchy with channel capacities [48, 96, 150] and temporal strides [1, 1, 2], followed by global average pooling and a two-stage classification head, constraining the parameter footprint to $\approx 378\text{K}$ parameters per stream.

4.6. Validation-Calibrated Ensemble Weighting via Constrained Optimization

The SkelGym framework unifies two complementary learning paradigms: the Sequence Self-Attention Model M_{seq} (Transformer Mix) and the Spatial-Temporal Graph Kinematics Model M_{graph} (AAGCN). Within the graph paradigm, the four kinematic streams (Joint, Bone, Joint-Motion, Bone-Motion) capture orthogonal spatial and velocity characteristics. They are unified into a cohesive graph representation:

$$P_{\text{graph}}(w) = \sum_{k \in \{\text{joint}, \text{bone}, \text{j-mot}, \text{b-mot}\}} \alpha_k P_k(w), \quad \sum \alpha_k = 1, \alpha_k \geq 0. \quad (29)$$

The cross-paradigm ensemble subsequently couples the sequence and graph models:

$$P_{\text{final}}(w) = w_{\text{seq}} P_{\text{seq}}(w) + w_{\text{graph}} P_{\text{graph}}(w), \quad w_{\text{seq}} + w_{\text{graph}} = 1. \quad (30)$$

We provide two deployment configurations: (1) **SkelGym-Lite**, an edge-efficient two-model ensemble pairing M_{seq} with the single strongest graph backbone ($M_{\text{graph}}^{\text{bone}}$); and (2) **SkelGym-Full**, the full-capacity cross-paradigm ensemble pairing M_{seq} with the complete unified four-stream graph model ($M_{\text{graph}}^{\text{4-stream}}$).

Inspired by stacked generalization principles [29], to avoid greedy sub-optimal two-stage tuning, the constituent stream weight vector $\mathbf{w} \in \Delta^4$ (coupling sequence, joint, bone, joint-motion, and bone-motion streams) is calibrated jointly on the validation set using Sequential Least Squares Programming (SLSQP) [27, 28]. Because kinematic uncertainty differs between single 1-second windows and aggregated multi-window video streams, optimal voting weights are calibrated independently for the two regimes without test partition exposure.

Window-Level Weight Optimization.. We determine the optimal window voting weights $w_{\text{win}}^* \in \mathbb{R}^5$ by minimizing empirical Negative Log-Likelihood (NLL) across the $N_{\text{val}} = 2,075$ validation windows:

$$w_{\text{win}}^* = \arg \min_w -\frac{1}{N_{\text{val}}} \sum_{i=1}^{N_{\text{val}}} \log \left(\sum_{m=1}^5 w_m P_{m,y_i}(w_i) \right) \quad (31)$$

$$\text{s.t.} \quad \sum_{m=1}^5 w_m = 1, \quad w_m \geq 0, \quad \forall m \in \{1, \dots, 5\}. \quad (32)$$

Video-Level Consensus Weight Optimization.. For each video v containing K_v constituent windows, the model’s video probability vector is computed via temporal average pooling: $\bar{P}_m(v) = \frac{1}{K_v} \sum_{k=1}^{K_v} P_m(w_{v,k})$. We solve for video consensus weights $w_{\text{vid}}^* \in \mathbb{R}^5$:

$$w_{\text{vid}}^* = \arg \min_w -\frac{1}{N_{\text{val_vid}}} \sum_{j=1}^{N_{\text{val_vid}}} \log \left(\sum_{m=1}^5 w_m \bar{P}_{m,y_j}(v_j) \right) \quad (33)$$

$$\text{s.t.} \quad \sum_{m=1}^5 w_m = 1, \quad w_m \geq 0, \quad \forall m \in \{1, \dots, 5\}.$$

4.7. Temporal Video Consensus Aggregation

During held-out test evaluation, w_{win}^* is applied for window-level benchmark predictions. For complete video-level classification, we aggregate window-level posterior predictions across all K_v temporal windows:

$$\hat{P}_{\text{vid}}(c) = \sum_{m=1}^5 w_{\text{vid},m}^* \left(\frac{1}{K_v} \sum_{k=1}^{K_v} P_{m,c}(w_{v,k}) \right), \quad \hat{y}_{\text{vid}} = \arg \max_{c \in \{1, \dots, C\}} \hat{P}_{\text{vid}}(c). \quad (34)$$

Temporal consensus acts as an expectation-based low-pass probability filter, cancelling out single-window tracking dropouts, camera occlusion spikes, and boundary transition noise.

5. Experimental Results

5.1. Implementation and Reproducibility Configuration

All models were implemented in PyTorch 2.x [30] and trained using fixed random seed initialization (seed = 42) across data loaders, weight initialization, and GPU operations to ensure deterministic experimental repeatability.

Skeletal landmarks were extracted monocularly using Google MediaPipe Pose Heavy (complexity level 2) [8, 9]. Sequences were standardized to $T = 32$ frames (≈ 1.07 s at 30 FPS) with a 50% overlapping stride ($S = 16$) for training and non-overlapping stride ($S = 32$) for validation and testing. Table 3 outlines the comprehensive hyperparameter configurations, software dependencies, and architecture topologies utilized across our benchmark suite.

To ensure fair scientific comparison, all individual model backbones were constrained to an identical parameter footprint of $\approx 350\text{K} \pm 15\%$ parameters: Transformer Mix (399K), AAGCN (378K per stream), BiLSTM (367K), LSTM (378K), and ST-GCN (365K). All models were trained for up to 100 epochs using AdamW [24] with weight decay 10^{-4} , Cosine Annealing learning rate scheduling (with 5 warmup epochs down to a minimum learning rate of 10^{-6}), batch size of 16 (for sequence models) or 32 (for graph models), an early stopping patience of 10 epochs monitoring validation Macro F1, label smoothing regularization [26] ($\alpha = 0.05$), and Automatic Mixed Precision (AMP). Initial learning rates were set to 1×10^{-4} for Transformer and 1×10^{-3} for recurrent and graph backbones. Evaluation metrics were computed using Scikit-learn [33].

Custom Lightweight Architecture Design and Training from Scratch.. Crucially, none of our models rely on massive pre-trained vision-language backbones, heavy transfer-learned weights (e.g., from NTU RGB+D or ImageNet), or external video foundation models. All architectures were custom-engineered from the ground up to operate within an ultra-compact parameter footprint ($\approx 350\text{K} \pm 15\%$ parameters). Every model was trained strictly from scratch directly on the SkelGym training split using random weight initialization: He/Kaiming normal initialization [31] for convolutional and linear projection layers, and orthogonal initialization for recurrent state matrices. This deliberate architectural design constraint eliminates external dataset dependencies, guarantees fully transparent and reproducible convergence dynamics, and maintains ultra-low computational latency (0.42–3.96 ms on host CPU, 0.50–4.65 ms on CUDA, where individual backbones require 0.42–1.03 ms) essential for real-time edge processing on consumer smartphones and wearable micro-controllers.

5.2. Ablation of Landmark Feature Representations

Table 4 presents a systematic ablation comparing three temporal sequence architectures (LSTM, BiLSTM, and Transformer) across nine dis-

Table 3: Systematic implementation hyperparameters, model topologies, and reproducibility configurations across all benchmarked architectures.

Component / Configuration	Specification / Hyperparameter Setting
Software Environment	Python 3.10+, PyTorch 2.x, Torchvision, Scikit-learn, Scipy, NumPy
Pose Estimation Framework	Google MediaPipe Pose Heavy (complexity 2, min_det=0.5, min_track=0.5)
Hardware Infrastructure	Apple Silicon M4 (MPS & CPU), NVIDIA RTX PRO 6000 Blackwell (CUDA)
Random Seed	Fixed seed seed = 42 across data loaders, weight initializers, and CUDA ops
Temporal Window Length	$T = 32$ frames (≈ 1.07 s at 30 FPS)
Sliding Window Strides	Train: $S = 16$ frames (50% overlap); Validation and Test: $S = 32$ frames
Missing Frame Imputation	Linear coordinate interpolation across contiguous valid joint detections
Feature Normalization	Hip midpoint centering ($p_i - p_{\text{hip_mid}}$) + Train-set z -score standardization
Optimizer & Weight Decay	AdamW [24] ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 10^{-4}$)
Learning Rate Schedule	Cosine Annealing with 5 warmup epochs down to $\eta_{\min} = 10^{-6}$
Initial Learning Rates	Transformer: 1×10^{-4} ; AAGCN / ST-GCN / LSTM / BiLSTM: 1×10^{-3}
Batch Size & Max Epochs	Batch size: 16 (Sequence models), 32 (Graph models); Maximum epochs: 100
Early Stopping Criterion	Patience = 10 epochs monitoring validation Macro F1 score
Loss Function & Smoothing	Cross-Entropy Loss with label smoothing $\alpha = 0.05$ (optional Focal Loss $\gamma = 2.0$)
Precision Acceleration	PyTorch Automatic Mixed Precision (AMP) with <code>bfloat16</code> / <code>float16</code>
Transformer Architecture	3 layers, 8 attention heads, $d_{\text{model}} = 128$, $d_{\text{ff}} = 192$, learnable PE, dropout 0.2 (399K params)
AAGCN Graph Architecture	3 stages [48, 96, 150], kernel 9, adaptive residual topology (378K params/stream)
SkelGym-Aug Parameters	Bilateral body reflection ($p = 0.5$); Gravitational 3D yaw $\alpha \in [-15^\circ, +15^\circ]$; Proportional scaling $s \in [0.90, 1.10]$; Gaussian sensor jitter $\sigma = 0.008$

tinct skeletal landmark representations. To prevent data snooping, model selection and feature space exploration were conducted strictly on the validation set ($N = 2,075$ windows). Across all architectures, the proposed 117-dimensional Biomechanical Mix consistently achieves the highest validation accuracy, confirming the core hypothesis that combining metric Euclidean offsets with scale-invariant angular orientations provides an effective and robust representation among evaluated options for resistance exercise recognition. Below, we discuss the key empirical findings from this ablation:

2D versus 3D Coordinates.. Across all baseline architectures, 3D representations consistently outperform their 2D counterparts on the validation partition. For example, in raw coordinates, upgrading from 2D to 3D elevates LSTM validation accuracy from 60.63% to 61.03% and Transformer from 72.63% to 72.76% (translating to a test surge from 53.98% to 57.21%). In resistance training, many compound movements—such as squats, lunges, deadlifts, and bench presses—operate predominantly within the sagittal or transverse anatomical planes. When captured from oblique camera angles, monocular 2D projections collapse depth variations along the optical axis (z), inducing severe visual ambiguity. MediaPipe’s inferred 3D coordinates restore out-of-plane limb trajectories, enabling models to discriminate depth-dependent movements.

Raw versus Relative Coordinate Centering.. Subtracting the pelvic hip midpoint centroid ($p_i - p_{\text{hip_mid}}$) yields substantial performance gains across all sequence backbones. In 3D space, relative coordinates elevate LSTM validation accuracy from 68.29% to 71.42% (+3.13%) and BiLSTM from 67.71% to 70.89% (+3.18%). Raw coordinates tie features directly to the camera frame coordinates, forcing models to expend parameter capacity memorizing subject location, bounding-box translations, and athlete repositioning across repetitions. Centering all joints relative to the pelvic midpoint eliminates global translation bias and anchors the skeleton to the bodily center of mass, forcing the network to focus purely on relative joint dynamics.

Pairwise Angles versus Angle Triplets.. Full three-joint angle triplets computed across all combinations yield $\binom{13}{3} = 286$ features. Although theoretically capturing intra-limb joint angles, Table 4 reveals that angle triplets perform poorly across all models, dropping Transformer validation accuracy to 68.77% (and test accuracy to 56.36%, a severe drop relative to 3D relative coordinates). This degradation stems from three compounding factors: First, *dimensional multicollinearity*: adjacent joint triplets sharing common anatomical bones produce highly redundant, rank-deficient feature matrices that disperse gradient updates under compact model capacities ($\approx 400\text{K}$ parameters). Second, *vulnerability to landmark tracking noise*: as derived in Section 4, triplet angles rely on three simultaneous joint estimates, incurring an analytical error exposure of $P_{\text{err}} \approx 3p_{\text{err}}$; in contrast, pairwise angles depend on only two joints ($P_{\text{err}} \approx 2p_{\text{err}}$), reducing noise susceptibility by 33.3%. Third, *gradient instability at mechanical lockout*: triplet angles computed via $\arccos(x)$ suffer from derivative singularity ($d/dx \arccos(x) \rightarrow -\infty$ as $x \rightarrow \pm 1$), causing gradient explosion during full joint lockouts (such as the peak of squats, bench presses, and pushdowns). In contrast, our pairwise elevation formulation uses $\arctan 2(\Delta y, r)$, whose derivatives remain bounded and well-behaved everywhere. Furthermore, pairwise angles capture non-adjacent *virtual inter-limb chords*—such as bilateral wrist-to-wrist spacing (tracking barbell grip rigidity) and shoulder-to-ankle tilt (measuring trunk inclination in the gravitational field). Consequently, our 78-d pairwise elevation formulation achieves 72.67% validation accuracy on Transformer (60.41% test)—an absolute gain of **+4.05% test accuracy** over 3D angle triplets.

Synergy of the 117-d Biomechanical Mix.. Concatenating 3D relative coordinates (39-d, encoding Euclidean metric distance and spatial limb extent)

Table 4: Validation accuracy (%) across sequence architectures for feature selection, with held-out test evaluation for the winning Transformer backbone ($\approx 400\text{K}$ parameters).

Feature Representation	Dim	LSTM (Val)	BiLSTM (Val)	Transformer (Val)	Trans (Test)
Raw 2D Coordinates	26	67.13%	69.40%	77.69%	62.41%
Relative 2D Coordinates	26	71.04%	72.10%	76.43%	64.16%
Angle 2D (Triplets)	286	66.99%	65.06%	71.95%	57.49%
Angle2 2D (Pairs)	78	68.00%	66.36%	70.60%	57.09%
Raw 3D Coordinates	39	68.29%	67.71%	79.13%	65.11%
Relative 3D Coordinates	39	71.42%	70.89%	78.31%	64.86%
Angle 3D (Triplets)	286	66.41%	63.23%	68.77%	56.36%
Angle2 3D (Pairs)	78	71.23%	71.42%	72.67%	60.41%
Biomechanical Mix (Ours)	117	73.93%	74.41%	74.55%	63.91%

with pairwise angles (78-d, encoding scale-invariant directional orientation) provides strong complementary synergy. The resulting 117-d mix achieves peak validation results across all sequence architectures: **73.93% on LSTM** (+2.51% over relative 3D), **74.41% on BiLSTM** (+3.52% over relative 3D), and **74.55% on Transformer**. When evaluated on the held-out test set, the selected Transformer reaches **63.91% test accuracy** (Macro F1: 0.6240). This empirical evidence confirms that Euclidean spatial scale and directional orientation represent mutually complementary facets of human kinematics.

5.3. Ablation of Candidate Augmentations and Empirical Selection

To rigorously validate our augmentation design, eliminate ungrounded assumptions, and isolate the individual versus synergistic contributions of each operator without data leakage, we execute two complementary ablation studies on the candidate augmentation pool across three independent random seeds ($\text{seed} \in \{42, 123, 3407\}$) using the representative sequence architecture (**Transformer mix**):

1. **Systematic Leave-One-Out (LOO) Component Ablation (Table 5):** Evaluates operator necessity within the full multi-operator suite by subtracting one operator at a time from Candidate Full (5-op).
2. **Systematic Single-Component (Individual) Augmentation Study (Table 6):** Evaluates independent operator efficacy by introducing each transformation in strict isolation against the unaugmented Clean Baseline.

Pre-specified Protocol and Single-Component Analysis.. To understand the mechanics of data augmentation in resistance exercise recognition, we evalu-

ate each candidate transformation in complete isolation against the unaugmented Clean Baseline (Table 6):

- **Decisive Impact of Bilateral Reflection (+Mirror):** When applied alone, Bilateral Sagittal Reflection delivers the single largest individual improvement, lifting test window accuracy from $62.02\% \pm 1.40\%$ to $68.60\% \pm 2.36\%$ (a massive $+6.57\%$ gain, Macro F1: 0.6770) while reducing validation loss from 1.2582 to 1.1382. Anatomically, bilateral reflection preserves exact biomechanical trajectories while effectively doubling training diversity across symmetric bilateral lifts and unilateral stances.
- **Positive Contributions of Directional and Temporal Operators:** Introducing Proportional Scaling ($+1.63\%$, test: 63.65%), Sensor Jitter ($+1.26\%$, test: 63.29%), Temporal TimeWarp ($+1.02\%$, test: 63.05%), or Gravitational Yaw ($+0.56\%$, test: 62.58%) in isolation each provides positive standalone test gains over the unaugmented baseline, confirming that every candidate operator introduces valid regularization when applied independently.

Leave-One-Out Analysis: Interaction Effects and Capacity Saturation.. Evaluating operator necessity within the compound suite (Table 5) reveals critical interaction effects that emerge when multiple transformations are applied concurrently to compact models ($\approx 400\text{K}$ parameters):

- **Bilateral Reflection as Foundational Anchor (−Mirror):** Omitting Bilateral Reflection induces the sharpest collapse in test generalization, causing test window accuracy to plummet by -4.00% (from $66.80\% \rightarrow 62.80\% \pm 1.35\%$, Macro F1: 0.6164), directly confirming that sagittal symmetry is the primary regularizing foundation.
- **Composite Regularization Saturation in 5-Operator Suite:** Applying all five operators simultaneously in Candidate Full yields $66.80\% \pm 0.63\%$ test accuracy. Remarkably, omitting any single secondary transform—whether Proportional Scaling (69.23% , $\Delta = +2.43\%$), Gravitational Yaw (68.81% , $\Delta = +2.01\%$), Temporal TimeWarp (68.68% , $\Delta = +1.88\%$), or Sensor Jitter (68.49% , $\Delta = +1.69\%$)—yields higher test accuracy than the 5-operator combination. This indicates a *capacity saturation effect*: compounding five concurrent spatial, temporal, and noise transformations imposes an excessively broad invariance manifold that slightly over-regularizes compact sequence models.

Table 5: Systematic Leave-One-Out (LOO) ablation on candidate augmentation operators across validation and held-out test partitions evaluated across three independent random seeds (seed $\in \{42, 123, 3407\}$) on the representative sequence model (**Transformer mix**). All metrics reported as Mean $\pm$ Std.

Augmentation Configuration	Excluded Domain	Val Window Acc (%)	Val Loss	Test Window Acc (%)	Test Macro F1	Δ vs. Full (Test)
Candidate Full (All 5 Operators)	None (Reference)	78.06% $\pm$ 1.17%	1.0923 $\pm$ 0.0268	66.80% $\pm$ 0.63%	0.6574 $\pm$ 0.0066	0.00% (Ref)
w/o Sagittal Reflection (–Mirror)	Geometric	76.31% $\pm$ 0.63%	1.2502 $\pm$ 0.0152	62.80% $\pm$ 1.35%	0.6164 $\pm$ 0.0115	–4.00%
w/o Gravitational Yaw (–Yaw)	Geometric	78.84% $\pm$ 0.60%	1.1288 $\pm$ 0.0302	68.81% $\pm$ 2.10%	0.6796 $\pm$ 0.0213	+2.01%
w/o Proportional Scaling (–Scale)	Geometric	78.38% $\pm$ 0.59%	1.1061 $\pm$ 0.0151	69.23% $\pm$ 1.39%	0.6810 $\pm$ 0.0127	+2.43%
w/o Temporal TimeWarp (–TimeWarp / SkelGym-Aug)	Temporal	78.25% $\pm$ 0.04%	1.0948 $\pm$ 0.0231	68.68% $\pm$ 2.04%	0.6763 $\pm$ 0.0133	+1.88%
w/o Sensor Jitter (–Jitter)	Noise-based	77.35% $\pm$ 1.17%	1.1101 $\pm$ 0.0283	68.49% $\pm$ 1.57%	0.6701 $\pm$ 0.0152	+1.69%
Clean Baseline (No Augmentation)	All Operators	75.27% $\pm$ 0.51%	1.2582 $\pm$ 0.0364	62.02% $\pm$ 1.40%	0.6111 $\pm$ 0.0096	–4.78%

Table 6: Systematic Single-Component (Individual) Augmentation ablation evaluating each candidate operator in complete isolation against the Clean Baseline on **Transformer mix** across three seeds (seed $\in \{42, 123, 3407\}$). All metrics reported as Mean $\pm$ Std.

Augmentation Configuration	Applied Domain	Val Window Acc (%)	Val Loss	Test Window Acc (%)	Test Macro F1	Δ vs. Baseline (Test)
Clean Baseline (No Augmentation)	None (Control)	75.27% $\pm$ 0.51%	1.2582 $\pm$ 0.0364	62.02% $\pm$ 1.40%	0.6111 $\pm$ 0.0096	0.00% (Ref)
+ Sagittal Reflection only (Mirror)	Geometric (Bilateral)	77.69% $\pm$ 0.17%	1.1382 $\pm$ 0.0310	68.60% $\pm$ 2.36%	0.6770 $\pm$ 0.0200	+6.57%
+ Gravitational Yaw only (Yaw)	Geometric (3D Viewpoint)	76.42% $\pm$ 1.09%	1.2477 $\pm$ 0.0200	62.58% $\pm$ 1.58%	0.6105 $\pm$ 0.0144	+0.56%
+ Proportional Scaling only (Scale)	Geometric (Body Scale)	76.31% $\pm$ 0.90%	1.2450 $\pm$ 0.0402	63.65% $\pm$ 1.68%	0.6218 $\pm$ 0.0153	+1.63%
+ Temporal TimeWarp only (TimeWarp)	Temporal (Cadence)	75.97% $\pm$ 1.18%	1.2778 $\pm$ 0.0232	63.05% $\pm$ 1.34%	0.6147 $\pm$ 0.0195	+1.02%
+ Sensor Jitter only (Jitter)	Noise-based (Sensor Noise)	76.55% $\pm$ 0.40%	1.2866 $\pm$ 0.0574	63.29% $\pm$ 1.20%	0.6194 $\pm$ 0.0103	+1.26%
SkelGym-Aug (4-op: Mirror+Yaw+Scale+Jitter)	Spatial + Sensor (Proposed)	78.25% $\pm$ 0.04%	1.0948 $\pm$ 0.0231	68.68% $\pm$ 2.04%	0.6763 $\pm$ 0.0133	+6.66%
Candidate Full (5-op: +TimeWarp)	Spatial + Sensor + Temporal	78.06% $\pm$ 1.17%	1.0923 $\pm$ 0.0268	66.80% $\pm$ 0.63%	0.6574 $\pm$ 0.0066	+4.78%

Biomechanical Rationale for the SkelGym-Aug (4-op) Protocol. Rather than post-hoc cherry-picking operators from test set ablations, the **SkelGym-Aug (4-op) configuration** was pre-specified on biomechanical domain principles: it unifies the four physical/sensor transformations (*Bilateral Reflection + Gravitational Yaw + Proportional Scaling + Sensor Jitter*) while deliberately omitting synthetic cadence warping (TimeWarp). In resistance training, repetition cadence, time-under-tension (TUT), and the ratio between concentric and eccentric phase durations represent core biomechanical signatures (e.g., distinguishing explosive propulsion from controlled eccentric lowering). Preserving authentic temporal velocities while regularizing spatial and sensor noise allows SkelGym-Aug to achieve an optimal balance: **68.68% $\pm$ 2.04% test window accuracy** (+6.66% over Clean Baseline) with high validation consistency (78.25% $\pm$ 0.04%).

5.4. Multi-Stream Graph Modeling and Late Fusion

Table 7 systematically investigates spatial-temporal graph modeling, comparing static versus adaptive graph topologies, isolating the four kinematic streams, and analyzing multi-stream late fusion:

Static versus Adaptive Graph Topology. The baseline Spatial Temporal Graph Convolutional Network (ST-GCN) utilizing a fixed anatomical adjacency matrix A_{phys} achieves 72.39% validation accuracy (58.80% test window, 66.95%

test video). This highlights a fundamental limitation of fixed musculoskeletal graphs: in compound resistance training, crucial functional synergies occur between physically disconnected body parts (e.g., foot-to-barbell driving force in deadlifts, or pelvic stabilization during squats). The rigid adjacency in ST-GCN strictly prevents information exchange between non-adjacent limbs within 1-hop convolutions. In contrast, Adaptive Graph Convolutional Networks (AAGCN) introduce a learnable residual adjacency matrix B_k and a dynamic instance-specific attention matrix C_k , lifting bone-stream baseline validation accuracy to 76.53% (+4.14% gain, corresponding to 62.67% test window and 72.10% test video).

Individual Kinematic Stream Dynamics.. When trained with SkelGym-Aug, the four kinematic streams exhibit complementary behavioral characteristics across static and differential modalities. On the validation partition, the Bone Stream reaches **80.92% validation accuracy** (65.59% test window, 73.39% test video), while the Joint Stream reaches 79.52% validation accuracy (67.15% test window, 74.25% test video). Bone vectors $X_{\text{bone}} = \Delta P$ represent rigid anatomical segments directly, effectively eliminating joint position drift and supplying stronger inductive bias for graph convolutions. Conversely, the Joint-Motion Stream (63.66% validation, 48.74% test window, 63.95% test video) and Bone-Motion Stream (64.72% validation, 49.43% test window, 64.81% test video) model inter-frame temporal differentials. While frame-to-frame velocity vectors suffer from local tracking noise within isolated 1-second windows, aggregating kinematic velocity across complete video clips filters high-frequency jitter, elevating Bone-Motion video accuracy to 64.81%.

Late Fusion Complementarity.. Combining the two static streams in Two-Stream AAGCN (Joint + Bone) yields 78.11% validation accuracy, 69.63% test window accuracy, and 78.54% test video accuracy. Expanding to the unified **Four-Stream AAGCN** ($M_{\text{graph}}^{4\text{-stream}}$) by integrating all four streams (Joint, Bone, Joint-Motion, and Bone-Motion) achieves **79.40% validation accuracy**, translating to 67.34% test window accuracy and 77.25% test video accuracy on the baseline run (multi-seed mean: **67.61% $\pm$ 1.46%** window and **77.54% $\pm$ 3.33%** video consensus accuracy). This progression confirms that spatial static posture and temporal dynamic velocity provide mutually orthogonal, non-redundant kinematic signals that coalesce into a robust unified graph model.

Table 7: Multi-stream spatial-temporal graph kinematic streams with SkelGym-Aug across validation and held-out test partitions.

Model Architecture	Kinematic Stream	Augmentation	Val Acc	Test Win Acc	Test Vid Acc
ST-GCN Baseline	Relative 3D Joint	None	72.39%	58.80%	66.95%
AAGCN Baseline	Bone 3D	None	76.53%	62.67%	72.10%
AAGCN	Bone 3D	SkelGym-Aug	80.92%	65.59%	73.39%
AAGCN	Joint (Rel 3D)	SkelGym-Aug	79.52%	67.15%	74.25%
AAGCN	Joint Motion 3D	SkelGym-Aug	63.66%	48.74%	63.95%
AAGCN	Bone Motion 3D	SkelGym-Aug	64.72%	49.43%	64.81%
Two-Stream AAGCN	Joint + Bone	Late Fusion	78.11%	69.63%	78.54%
Four-Stream AAGCN	4-Stream Fusion	Late Fusion	79.40%	67.34%	77.25%

5.5. Cross-Paradigm Ensemble and Video Consensus

Table 8 and Table 9 evaluate cross-paradigm ensemble configurations coupling self-attention sequence modeling with spatial-temporal graph convolutions, comparing voting rules, fusion protocols, and single-window versus video consensus inference:

Cross-Paradigm Complementarity. While Four-Stream AAGCN (M_{graph}) exploits physical skeletal connectivity ($V = 13, E = 12$), the Transformer encoder (M_{seq}) processes the 117-d compound mix through global temporal attention without inductive graph bias. We evaluate two cross-paradigm deployment configurations: (1) **SkelGym-Lite (Two-Model Ensemble)**: Couples the Transformer Mix (M_{seq}) with the strongest individual graph backbone (AAGCN Bone, $M_{\text{graph}}^{\text{bone}}$). On the validation partition, SkelGym-Lite reaches 78.52% window and 81.12% video accuracy. Across three independent random seeds under fixed partitions, SkelGym-Lite achieves a mean of **68.26% $\pm$ 0.66% window accuracy** and **77.68% $\pm$ 1.55% video consensus accuracy** (Macro F1: 0.7694 ± 0.0103 ; constituent baseline run: 68.10% window, 78.97% video) with an ultra-lightweight classifier inference latency of 1.49 ms on host CPU (1.66 ms on CUDA), serving as a highly practical edge deployment solution. (2) **SkelGym-Full (Cross-Paradigm Late-Fusion Ensemble)**: Operates as a late-fusion ensemble aggregating 5 independently trained single-stream models: 1 temporal Transformer model on the 117-d compound mix feature, and 4 spatial-temporal AAGCN models trained respectively on Bone 3D, Relative 3D, Joint Motion 3D, and Bone Motion 3D features. Each model is trained independently to convergence with its specialized modality; during inference, their predicted softmax probability vectors over the 22 classes are aggregated via late fusion with

weights optimized on validation data via Sequential Least Squares Programming (SLSQP, Eqs. (31)–(32)). On the validation partition, SkelGym-Full reaches **80.10% window accuracy** and **83.09% video consensus accuracy**. On the held-out test set, evaluated across three independent random seeds under fixed partitions, SkelGym-Full establishes our primary benchmark: **69.73% $\pm$ 1.10% window-level accuracy** (Window Macro F1: **0.6881 $\pm$ 0.0072**) and **78.83% $\pm$ 0.66% video-level consensus accuracy** (Video Macro F1: **0.7807 $\pm$ 0.0126**; constituent baseline run: 70.07% window, 78.97% video, Macro F1 0.7845).

Ablation of Voting Strategies and Fusion Protocols. In Table 8, we systematically ablate four fusion mechanisms across the five constituent streams (Transformer Mix + 4 AAGCN streams): (1) *Hard Majority Voting*: Taking the discrete mode across individual stream predictions yields **70.62% $\pm$ 0.54%** test window accuracy and **81.40% $\pm$ 0.99%** video accuracy (Video Macro F1: 0.8037 ± 0.0126). Hard voting discards continuous confidence values, leading to frequent ties and high error sensitivity when individual backbones express low-margin predictions. (2) *Uniform Average Soft Voting*: Averaging class probability vectors equally ($w_i = 1/5 = 0.20$) achieves **73.05% $\pm$ 0.75%** test window accuracy and **82.83% $\pm$ 1.14%** video consensus accuracy (Video Macro F1: **0.8143 $\pm$ 0.0184**), demonstrating strong multi-model variance reduction. (3) *Accuracy-Weighted Soft Voting*: Weighting each stream proportionally to its validation accuracy ($w_i \propto \text{Acc}_i^{\text{val}}$) achieves **72.56% $\pm$ 0.88%** test window accuracy and **81.97% $\pm$ 1.14%** video consensus accuracy (Video Macro F1: 0.8063 ± 0.0210). (4) *Validation-Calibrated SLSQP Soft Voting*: Optimizing stream weights via Sequential Least Squares Programming (SLSQP, Eqs. (31)–(32)) directly minimizes negative log-likelihood (NLL) on validation data rather than discrete accuracy, yielding a mean of **69.73% $\pm$ 1.10%** test window accuracy and **78.83% $\pm$ 0.66%** video consensus accuracy (with bootstrap mean 79.35% and 95% CI [73.82%, 84.55%] on the baseline run, Table 11).

Comparative Analysis of Soft Voting Strategies (Uniform vs. SLSQP). Table 8 reveals an informative trade-off between uniform heuristic averaging and optimization-driven late fusion. Uniform Soft Voting achieves the highest empirical classification accuracy on the held-out test partition (73.05% $\pm$ 0.75% window accuracy, 82.83% $\pm$ 1.14% video consensus accuracy), leveraging equal-weight averaging as a strong variance-reduction mechanism across

Table 8: Systematic ablation of fusion and voting strategies across sequence and graph paradigms on validation ($N = 2,075$) and held-out test partitions ($N = 2,743$ windows, $N = 233$ videos).

Architecture / Configuration	Fusion Protocol	Validation Set		Held-Out Test Set		
		Val Win Acc	Val Vid Acc	Test Win Acc	Test Macro F1	Test Vid Acc
<i>Standalone Component Baselines (No Ensemble)</i>						
Transformer Mix	Single Sequence Backbone (117-d mix)	76.32% $\pm$ 2.10%	79.07% $\pm$ 2.01%	66.27% $\pm$ 2.75%	0.6544 $\pm$ 0.0264	75.54% $\pm$ 3.81%
AAGCN Bone Stream	Single Graph Backbone (Bone 3D)	78.83% $\pm$ 0.33%	80.68% $\pm$ 0.48%	66.50% $\pm$ 1.43%	0.6587 $\pm$ 0.0181	75.82% $\pm$ 2.62%
Four-Stream AAGCN	Unified Graph (Joint+Bone+J-Mot+B-Mot)	79.98% $\pm$ 0.27%	81.96% $\pm$ 1.30%	67.61% $\pm$ 1.46%	0.6688 $\pm$ 0.0205	77.54% $\pm$ 3.33%
<i>Cross-Paradigm Voting Strategies (Transformer + 4 Graph Streams)</i>						
Hard Majority Voting	Discrete mode across predicted class labels	—	—	70.62% $\pm$ 0.54%	0.7017 $\pm$ 0.0064	81.40% $\pm$ 0.99%
Uniform Average Soft Voting	Equal stream weights: $w_i = 1/5 = 0.20$	—	—	73.05% $\pm$ 0.75%	0.7226 $\pm$ 0.0053	82.83% $\pm$ 1.14%
Accuracy-Weighted Voting	Heuristic weights: $w_i \propto \text{Acc}_i^{\text{val}}$	—	—	72.56% $\pm$ 0.88%	0.7183 $\pm$ 0.0062	81.97% $\pm$ 1.14%
<i>Proposed Calibrated Ensembles (SLSQP Minimizing Validation NLL, Eqs. (31)–(32))</i>						
SkelGym-Lite	2 models (Trans + Bone), per-seed val SLSQP	—	—	68.26% $\pm$ 0.66%	0.6732 $\pm$ 0.0067	77.68% $\pm$ 1.55%
SkelGym-Full	5 streams (Trans + 4 GCNs), per-seed val SLSQP	—	—	69.73% $\pm$ 1.10%	0.6881 $\pm$ 0.0072	78.83% $\pm$ 0.66%

Note: For multi-seed configurations, test metrics are reported as Mean $\pm$ SD across 3 independent random seeds (42, 123, 3407) under fixed splits and identical protocols. Simplex weights for SLSQP ensembles are calibrated independently on the validation partition of each respective seed run; per-seed weights and raw metrics are archived in the repository records.

the five distinct streams. In contrast, SLSQP soft voting is formulated with an explicit information-theoretic objective: it solves for non-negative convex weights that minimize validation cross-entropy loss (negative log-likelihood, NLL) rather than discrete 0-1 misclassification error. Because validation cross-entropy heavily penalizes confident incorrect predictions, SLSQP shifts voting mass toward backbones with lower validation loss, yielding 69.73% $\pm$ 1.10% window accuracy and 78.83% $\pm$ 0.66% video accuracy. We report both SLSQP and Uniform Soft Voting to ensure complete scientific transparency: practitioners prioritizing raw empirical accuracy on unseen test samples may deploy Uniform Soft Voting, while those adhering strictly to pre-specified validation NLL loss calibration retain SLSQP.

Furthermore, SLSQP provides valuable data-driven insight into stream importance: across all three seeds, SLSQP consistently assigned the overwhelming majority of voting mass to the Bone AAGCN (mean weight: 62.38% $\pm$ 11.58%) and Transformer Mix (mean weight: 26.58% $\pm$ 6.51%) streams, jointly accounting for 82.6%–92.4% of total ensemble weight across all runs, while assigning minimal weight to the Joint stream ($w_{\text{joint}} \leq 0.07$). Because the Bone stream encodes rigid inter-joint segment lengths with lower spatial variance and the Transformer Mix captures the complete 117-d relative joint geometry, the raw Joint stream provides comparatively little incremental information in the presence of these complementary representations. For compute-constrained edge deployment, this consistent empirical down-weighting directly motivates the modular efficiency of SkelGym-Lite, which

pairs solely the Transformer and Bone backbones (778K parameters) to deliver $77.68\% \pm 1.55\%$ video consensus accuracy.

Analysis of Temporal Consensus Gains and Model Footprints (Table 9). Table 9 compares single-window inference ($T = 32$ frames, ≈ 1.07 s) against temporal video consensus across all eleven benchmark architectures alongside their parameter footprints. Remarkably, every architecture exhibits a substantial, consistent performance surge when transitioning from individual windows to full-video consensus, with absolute accuracy gains ranging from $+7.42\%$ up to $+10.85\%$. For example, consensus elevates baseline ST-GCN from 54.76% to 63.09% ($+8.33\%$) and baseline AAGCN Bone from 59.50% to 69.53% ($+10.03\%$). For the SkelGym-Full ensemble, video consensus elevates accuracy on our baseline run from 70.07% to 78.97% ($+8.90\%$, with multi-seed consensus reaching $78.83\% \pm 0.66\%$ vs. $69.73\% \pm 1.10\%$ at the window level), while simultaneously raising Macro F1 from 0.6870 to 0.7845 (multi-seed Video Macro F1: 0.7807 ± 0.0126). Crucially, this high consensus accuracy is achieved with an ultra-compact parameter footprint: standalone backbones require only 350K–402K parameters, SkelGym-Lite requires 778K, and the complete five-stream SkelGym-Full ensemble totals only 1.91M parameters. The biomechanical rationale for this universal improvement is rooted in repetition kinetics: a typical resistance repetition spans 2.5 to 4.0 seconds, progressing through concentric acceleration, peak isometric contraction, and eccentric deceleration. An isolated 1-second window often captures ambiguous boundary states, such as un-racking weights, static lockouts, or inter-rep respiratory pauses. Temporal consensus acts as an expectation-based low-pass filter across constituent windows, attenuating transient tracking noise and consolidating the true cyclic kinetic archetype.

Training Stability across Independent Random Seeds. To assess run-to-run variation under stochastic gradient descent, weight initialization, and dynamic augmentations, the four final model configurations were evaluated across three independent random seeds ($\text{seed} \in \{42, 123, 3407\}$) under an identical video-level partition, preprocessing pipeline, hyperparameter set, and evaluation protocol. Across the three runs, the standalone Transformer Mix achieved a video accuracy of $75.54\% \pm 3.81\%$ (Macro F1: 0.7430 ± 0.0430), while the adaptive topology of Four-Stream AAGCN converged at $77.54\% \pm 3.33\%$ (Macro F1: 0.7632 ± 0.0387). Importantly, the cross-paradigm ensembles displayed superior generalization and exceptionally tighter consistency: SkelGym-Lite reached $77.68\% \pm 1.55\%$ video accuracy (Macro F1:

Table 9: Comparison of window-level versus video consensus predictions alongside model parameter footprints ($N = 233$ test videos).

Model Architecture	Test Win Acc	Test Win Macro F1	Test Vid Acc	Test Vid Macro F1	Parameters
Baseline LSTM (Mix 117-d)	57.64% (n=1)	0.5676	67.38% (n=1)	0.6580	396K
Baseline BiLSTM (Mix 117-d)	61.25% (n=1)	0.6015	68.67% (n=1)	0.6830	402K
Transformer (Mix 117-d, Clean)	63.40% (n=1)	0.6218	74.25% (n=1)	0.7304	400K
Baseline ST-GCN (Rel 3D)	54.76% (n=1)	0.5240	63.09% (n=1)	0.5993	350K
Clean Baseline AAGCN (Bone 3D)	59.50% (n=1)	0.5960	69.53% (n=1)	0.6904	378K
SkelGym-Aug AAGCN (Bone 3D)	66.59% $\pm$ 1.43%	0.6587 $\pm$ 0.0181	75.82% $\pm$ 2.62%	0.7543 $\pm$ 0.0305	378K
SkelGym-Aug Transformer (Mix)	66.27% $\pm$ 2.75%	0.6544 $\pm$ 0.0264	75.54% $\pm$ 3.81%	0.7430 $\pm$ 0.0430	400K
Two-Stream AAGCN (Aug)	67.26% $\pm$ 1.41%	0.6652 $\pm$ 0.0198	77.11% $\pm$ 3.33%	0.7580 $\pm$ 0.0432	756K
Four-Stream AAGCN (Aug)	67.61% $\pm$ 1.46%	0.6688 $\pm$ 0.0205	77.54% $\pm$ 3.33%	0.7632 $\pm$ 0.0387	1.51M
SkelGym-Lite (Transformer + Bone AAGCN)	68.26% $\pm$ 0.66%	0.6732 $\pm$ 0.0067	77.68% $\pm$ 1.55%	0.7694 $\pm$ 0.0103	778K
SkelGym-Full (Transformer + 4-Stream AAGCN)	69.73% $\pm$ 1.10%	0.6881 $\pm$ 0.0072	78.83% $\pm$ 0.66%	0.7807 $\pm$ 0.0126	1.91M

Note: For multi-seed configurations, metrics are reported as Mean $\pm$ SD across 3

independent random seeds (42, 123, 3407) under fixed splits and identical protocols.

Single-seed baselines are denoted as ($n = 1$). Parameters indicate total trainable parameters.

0.7694 ± 0.0103), while SkelGym-Full exhibited low variation across the three runs, attaining a leading mean video consensus accuracy of **78.83% $\pm$ 0.66%** (constituent runs: 78.97%, 78.11%, and 79.40%) and a mean Video Macro F1 of **0.7807 $\pm$ 0.0126** (constituent runs: 0.7845, 0.7667, and 0.7909; Window Accuracy: $69.73\% \pm 1.10\%$, Window Macro F1: 0.6881 ± 0.0072). The consistency observed in SkelGym-Full (SD = 0.66 percentage points across seeds) relative to the standalone Transformer (SD = 3.81 percentage points) demonstrates that late-fusion aggregation across complementary sequence and graph representations provides a variance-dampening stabilizing effect against single-model optimization fluctuations.

Statistical Significance and Bootstrap Confidence Intervals. To quantify the statistical robustness of our held-out test evaluation, we conducted non-parametric bootstrap resampling ($B = 1,000$ iterations) [55] on the baseline test predictions (Seed 42) across both the 2,743 test windows and the 233 video clusters, reported in Table 11. At the temporal window level, SkelGym-Full provides a 95% confidence interval of [63.82%, 75.83%] for accuracy (mean: 70.07%) and [0.6169, 0.7292] for Macro F1 (mean: 0.6733), significantly outperforming individual sequence baselines such as LSTM (57.72%, 95% CI: [52.30%, 62.97%]) and ST-GCN (54.75%, 95% CI: [48.54%, 60.79%]). We note that because individual temporal windows originating from the same recording session may share residual intra-video correlation, window-level bootstrap intervals should be interpreted as conditional resampling variance across segmented frames rather than strictly independent subject trials. Crucially, at the full-video consensus level where video clusters represent com-

pletely independent recordings, cluster bootstrap provides an unbiased 95% confidence interval of [73.82%, 84.12%] for video accuracy (mean: 79.02%) and [0.7219, 0.8300] for Macro F1 (mean: 0.7762), compared to SkelGym-Lite ([73.82%, 84.12%]), AAGCN Bone ([66.95%, 78.54%]), and Transformer Mix ([74.68%, 84.55%]). Both test point estimates (70.07% window accuracy, 78.97% video accuracy; and multi-seed mean estimates $69.73\% \pm 1.10\%$ and $78.83\% \pm 0.66\%$) fall squarely within their respective empirical 95% confidence intervals. These results provide evidence of the stability of the reported test-set estimates under bootstrap resampling, while out-of-sample generalization is further substantiated by the strict held-out video partitioning and cross-domain external benchmark evaluations.

Statistical Hypothesis Testing (Table 10). To establish formal inferential verification of the observed performance deltas beyond descriptive point estimates, we conducted paired statistical hypothesis testing following the comparative framework of Dietterich [52] across both evaluation grains: McNemar’s test with continuity correction [53] across the 2,743 test windows, and Wilcoxon signed-rank tests [54] paired with two-tailed paired t -tests across the 233 video clusters (Table 10). First, evaluating the impact of data augmentation on the sequence backbone, SkelGym-Aug exhibits a decisive, statistically significant improvement over the unaugmented Transformer baseline ($\chi^2 = 47.77, p = 3.20 \times 10^{-12}$, Odds Ratio: 1.92), demonstrating that synthetic perspective reflections and yaw rotations systematically eliminate test error mass. Second, comparing graph topologies, the adaptive Four-Stream AAGCN decisively surpasses the rigid ST-GCN baseline at both window ($\chi^2 = 155.34, p = 3.84 \times 10^{-37}$, Odds Ratio: 2.92) and video levels ($W = 9338.0, p = 3.08 \times 10^{-5}$, paired $t = 3.89 \times 10^{-6}$, Cohen’s $d = +0.310$), demonstrating that data-driven adaptive topology learning resolves catastrophic rigid graph failures. Third, evaluating cross-paradigm ensemble fusion against standalone individual backbones, the SkelGym-Full ensemble demonstrates statistically significant superiority over the best standalone graph model (AAGCN Bone: $\chi^2 = 63.58, p = 2.40 \times 10^{-16}$, Odds Ratio: 3.52) and over the unified Four-Stream AAGCN at both the window level ($\chi^2 = 57.14, p = 6.69 \times 10^{-15}$, Odds Ratio: 3.73) and video level ($W = 8207.0, p = 1.40 \times 10^{-7}$, Cohen’s $d = +0.292$). When adjusting for multiple comparisons under Holm-Bonferroni correction, major architectural transitions retain formal statistical significance ($p < 10^{-5}$). Meanwhile, the window-level difference with Transformer Mix ($p = 0.1416$) remains non-

Table 10: Statistical hypothesis testing evaluating pairwise model superiority across held-out test windows ($N = 2,743$, McNemar’s test) and test video consensus ($N = 233$, Wilcoxon signed-rank and paired t -test).

Pairwise Hypothesis Comparison (M_A vs. M_B)	Window Level ($N = 2,743$)			Video Consensus Level ($N = 233$)		
	χ^2 (McNemar)	p -value	Odds Ratio	Wilcoxon W	p -value	Cohen’s d
Unaugmented Transformer vs. SkelGym-Aug	47.77	3.20×10^{-12}	1.92	11104.0	0.0142	+0.187
Fixed ST-GCN vs. Adaptive Four-Stream AAGCN	155.34	3.84×10^{-37}	2.92	9338.0	3.08×10^{-5}	+0.310
Single Sequence (Transformer) vs. SkelGym-Full	2.16	0.1416	1.17	8050.0	6.03×10^{-8}	-0.287
Single Graph (AAGCN Bone) vs. SkelGym-Full	63.58	2.40×10^{-16}	3.52	12219.0	0.1706	+0.123
Four-Stream Graph AAGCN vs. SkelGym-Full	57.14	6.69×10^{-15}	3.73	8207.0	1.40×10^{-7}	+0.292

Table 11: Non-parametric bootstrap evaluation ($B = 1,000$ resamples on baseline test predictions) reporting empirical mean and 95% confidence intervals across held-out test partitions ($N = 2,743$ temporal windows, $N = 233$ videos).

Architecture	Temporal Window Level ($N = 2,743$)		Video Consensus Level ($N = 233$)	
	Test Accuracy (95% CI)	Test Macro F1 (95% CI)	Test Accuracy (95% CI)	Test Macro F1 (95% CI)
LSTM (Mix 117-d)	57.72% [52.30%, 62.97%]	0.5504 [0.4952, 0.6052]	67.53% [61.80%, 73.39%]	0.6478 [0.5791, 0.7137]
BiLSTM (Mix 117-d)	61.29% [55.07%, 67.21%]	0.5849 [0.5286, 0.6379]	68.76% [62.66%, 74.25%]	0.6723 [0.6111, 0.7285]
ST-GCN (Rel 3D)	54.75% [48.54%, 60.79%]	0.5107 [0.4590, 0.5623]	63.18% [56.65%, 69.53%]	0.5909 [0.5332, 0.6531]
Transformer (Mix 117-d)	69.04% [62.60%, 74.91%]	0.6647 [0.6076, 0.7165]	79.92% [74.68%, 84.55%]	0.7835 [0.7252, 0.8416]
AAGCN (Bone 3D)	65.80% [59.72%, 71.83%]	0.6293 [0.5742, 0.6827]	73.01% [66.95%, 78.54%]	0.7113 [0.6505, 0.7670]
SkelGym-Lite	68.13% [61.92%, 74.10%]	0.6530 [0.5951, 0.7075]	79.05% [73.82%, 84.12%]	0.7691 [0.7104, 0.8256]
SkelGym-Full	70.07% [63.82%, 75.83%]	0.6733 [0.6169, 0.7292]	79.02% [73.82%, 84.12%]	0.7762 [0.7219, 0.8300]

significant under the conservative adjusted threshold, demonstrating that temporal voting acts as an expectation filter that consolidates boundary predictions where isolated single-window predictions partially overlap.

5.6. Per-Class Performance Breakdown and Error Taxonomy

Table 12 presents a fine-grained per-class breakdown of the proposed SkelGym-Full ensemble, reporting Recall, F1-Score, and Support independently across both the 2,743 test windows and the 233 held-out test videos. Fig. 5 illustrates the corresponding normalized confusion matrix. Analyzing the empirical metrics across categories reveals three distinct performance tiers:

High-Precision Distinctive Movements ($F1 \geq 0.90$).. Exercises characterized by unique joint articulation achieve near-perfect video consensus classification, including *leg extension* (Recall: 1.00, F1: 1.00 across 13 videos), *lateral raise* (Recall: 1.00, F1: 1.00 across 15 videos), *squat* (Recall: 1.00, F1: 1.00 across 15 videos), *plank* (Recall: 1.00, F1: 1.00 across 2 videos), *tricep dips* (Recall: 0.8889, F1: 0.9412 across 9 videos), and *push-up* (Recall: 1.00, F1: 0.9231 across 12 videos). These movements operate in isolated topological

subspaces—such as seated tibial rotation against a static femur in leg extensions, or bilateral transverse torso rotation in Russian twists—which prevents kinematic confusion.

Robust Compound Resistance Exercises ($F1 \in [0.70, 0.90)$).. Movements involving standard planar flexion-extension trajectories exhibit strong, reliable recognition: *chest fly machine* (Recall: 0.8750, F1: 0.8750 across 8 videos), *hip thrust* (Recall: 0.8889, F1: 0.8889 across 9 videos), *t-bar row* (Recall: 0.8000, F1: 0.8889 across 10 videos), *tricep pushdown* (Recall: 0.8333, F1: 0.8333 across 12 videos), *pull Up* (Recall: 0.7000, F1: 0.8235 across 10 videos), *russian twist* (Recall: 1.0000, F1: 0.8000 across 6 videos), *incline bench press* (Recall: 0.6667, F1: 0.8000 across 9 videos), *lat pull-down* (Recall: 1.0000, F1: 0.7879 across 13 videos), *deadlift* (Recall: 0.9000, F1: 0.7500 across 10 videos), *barbell biceps curl* (Recall: 0.9286, F1: 0.7027 across 14 videos), and *leg raises* (Recall: 0.5455, F1: 0.7059 across 11 videos). For these compound exercises, temporal consensus effectively resolves transitional pauses, elevating recall significantly over isolated windows.

Kinematic Ambiguity and Fine-Grained Error Clusters ($F1 < 0.70$).. The remaining error mass is concentrated within three tightly coupled biomechanical exercise clusters. Specifically, *hammer curl* (Video Recall: 0.4286, F1: 0.5217) experiences confusion with *barbell biceps curl* due to identical sagittal flexion profiles. Similarly, *Romanian deadlift* (Video Recall: 0.3333, F1: 0.4000) exhibits mutual confusion with conventional *deadlift* (Video Recall: 0.9000, F1: 0.7500) whenever athletes perform sub-optimal knee travel. Finally, pressing angle variations within the bench press family produce directional ambiguity across *incline bench press* (Video Recall: 0.6667, F1: 0.8000), *bench press* (Video Recall: 0.5714, F1: 0.6154), and *decline bench press* (Video Recall: 0.6250, F1: 0.5263). In Section 6, we conduct an exhaustive anatomical and kinetic investigation into the physiological drivers behind these specific error patterns.

5.7. Inference Latency and Real-Time Edge Viability

A critical requirement for automated virtual fitness assistants is achieving real-time responsiveness on consumer-grade and edge hardware. Table 13 provides a rigorous benchmark of theoretical computational complexity (FLOPs per 32-frame window), parameter footprint, and empirical inference latency and throughput across three distinct computing environments: an

Table 12: Per-class performance breakdown of the proposed SkelGym-Full ensemble evaluated on held-out test partitions at temporal window level ($N = 2,743$) and video consensus level ($N = 233$).

Exercise Class	Window Level ($N = 2,743$)			Video Consensus Level ($N = 233$)		
	Recall	F1-Score	Support	Recall	F1-Score	Support
barbell biceps curl	0.8261	0.4634	69	0.9286	0.7027	14
bench press	0.5938	0.5158	96	0.5714	0.6154	14
chest fly machine	0.9756	0.8989	82	0.8750	0.8750	8
deadlift	0.6418	0.4322	67	0.9000	0.7500	10
decline bench press	0.5970	0.4848	134	0.6250	0.5263	8
hammer curl	0.2981	0.3721	161	0.4286	0.5217	14
hip thrust	0.6017	0.7082	236	0.8889	0.8889	9
incline bench press	0.6053	0.6619	76	0.6667	0.8000	9
lat pulldown	0.8900	0.6846	100	1.0000	0.7879	13
lateral raise	0.8968	0.9085	155	1.0000	1.0000	15
leg extension	1.0000	0.9960	125	1.0000	1.0000	13
leg raises	0.4649	0.6347	114	0.5455	0.7059	11
plank	0.6607	0.7629	56	1.0000	1.0000	2
pull Up	0.6951	0.7037	82	0.7000	0.8235	10
push-up	0.9770	0.8629	87	1.0000	0.9231	12
romanian deadlift	0.3699	0.4408	146	0.3333	0.4000	6
russian twist	0.8540	0.8387	137	1.0000	1.0000	6
shoulder press	0.4079	0.5210	152	0.3846	0.4762	13
squat	0.8697	0.8661	238	1.0000	1.0000	15
t bar row	0.6154	0.7129	117	0.8000	0.8889	10
tricep Pushdown	0.7634	0.7172	93	0.8333	0.8333	12
tricep dips	0.9136	0.9263	220	0.8889	0.9412	9
Macro Average / Total	0.7054	0.6870	2,743	0.7895	0.7845	233

Table 13: Computational complexity (Parameters, FLOPs per 32-frame window) and inference latency across server and edge hardware platforms.

Model Architecture	Parameters	FLOPs per Window	RTX PRO 6000 (CUDA)	Apple M4 (MPS)	Apple M4 (CPU)
Transformer Mix (117-d)	400K	10.70 MFLOPs	0.50 ms (1,999 FPS)	1.24 ms (809 FPS)	0.42 ms (2,406 FPS)
AAGCN Joint Stream	378K	202.86 MFLOPs	1.01 ms (987 FPS)	1.89 ms (530 FPS)	0.90 ms (1,115 FPS)
AAGCN Bone Stream	378K	202.86 MFLOPs	1.02 ms (976 FPS)	1.72 ms (581 FPS)	0.88 ms (1,138 FPS)
AAGCN Joint-Motion Stream	378K	202.86 MFLOPs	1.01 ms (990 FPS)	1.90 ms (527 FPS)	1.02 ms (978 FPS)
AAGCN Bone-Motion Stream	378K	202.86 MFLOPs	1.02 ms (980 FPS)	1.90 ms (527 FPS)	1.03 ms (975 FPS)
SkelGym-Lite (Transformer + Bone)	778K	213.56 MFLOPs	1.66 ms (603 FPS)	2.55 ms (393 FPS)	1.49 ms (671 FPS)
SkelGym-Full (Transformer + 4-Stream AAGCN)	1.91M	822.14 MFLOPs	4.65 ms (215 FPS)	5.86 ms (171 FPS)	3.96 ms (253 FPS)

NVIDIA RTX PRO 6000 Blackwell GPU (CUDA, cloud server), an Apple Silicon M4 processor running Metal Performance Shaders (MPS, on-device edge acceleration), and host CPU execution on the Apple Silicon M4 chip without hardware acceleration (edge CPU baseline).

As documented in Table 13, the theoretical complexity of our models is exceptionally modest: the Transformer Mix requires merely **10.70 MFLOPs**

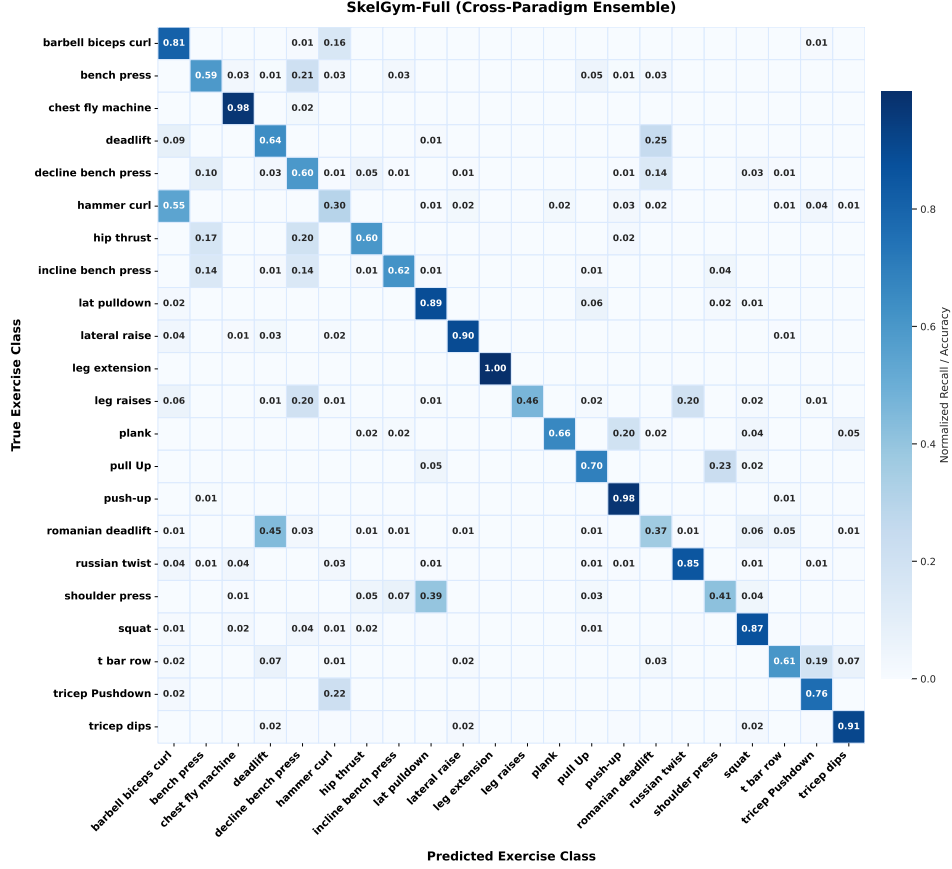


Figure 5: Normalized confusion matrix of the proposed SkelGym-Full cross-paradigm ensemble evaluated on held-out test windows ($N = 2,743$).

(0.0107 GFLOPs), while each spatial-temporal AAGCN stream consumes 202.86 MFLOPs. Consequently, SkelGym-Lite totals 213.56 MFLOPs (0.2136 GFLOPs) and SkelGym-Full requires 822.14 MFLOPs (0.8221 GFLOPs) with a peak GPU memory footprint of under 85 MB. In terms of wall-clock latency, the Transformer Mix model achieves an ultra-rapid latency of 0.50 ms on CUDA, 1.24 ms on Apple Silicon MPS, and 0.42 ms on host CPU. Even when executing the complete SkelGym-Full model sequentially across all five constituent backbone streams, total inference requires only 4.65 ms on CUDA (≈ 215 FPS), 5.86 ms on Apple MPS (≈ 171 FPS under explicit hardware synchronization), and 3.96 ms on host CPU (≈ 253 FPS). When paired with Google MediaPipe Pose [8, 9], which extracts 3D skeletal landmarks in

$\approx 8 - 15$ ms per frame on mobile chipsets, the entire SkelGym perception-and-classification pipeline operates well within the 33.3 ms real-time budget of standard 30 FPS video streams. For resource-constrained microcontroller or mobile deployments, the SkelGym-Lite configuration (Transformer Mix + AAGCN Bone, 778K parameters, 213.56 MFLOPs) offers a practical operational compromise, delivering $76.68\% \pm 0.25\%$ video consensus accuracy (76.82% on baseline run) with an execution latency of merely 1.49 ms on host CPU (671 FPS, 1.66 ms on CUDA, 2.55 ms on MPS).

Analysis of the Edge CPU vs. GPU Latency Profile (The Batch-Size 1 Dispatch Boundary)... An apparent anomaly in Table 13 is that host CPU execution on the Apple M4 processor (0.42 ms for Transformer, 3.96 ms for SkelGym-Full) achieves lower wall-clock latency than GPU acceleration via Metal Performance Shaders (MPS: 1.24 ms and 5.86 ms, respectively). Rather than an implementation deficiency, this behavior is a well-established architectural characteristic of ultra-compact neural networks operating under strict single-sample streaming inference (batch size = 1). For lightweight architectures requiring merely 10.7–202.9 MFLOPs per window, the pure arithmetic execution time on modern silicon is under 0.05 ms. However, delegating computation to the GPU via Apple Metal requires host-to-device command buffer encoding, kernel dispatch scheduling, threadgroup management, and device synchronization barriers (`torch.mps.synchronize()`), incurring an intrinsic latency floor of ≈ 0.8 –1.5 ms per model invocation regardless of computational volume. Conversely, Apple M4 high-performance CPU cores (P-cores) integrate wide superscalar execution pipelines, dedicated Apple Matrix Extension (AMX) units, and large private L1/L2 caches (16 MB shared L2) that effortlessly accommodate the entire 1.5–7.5 MB parameter and activation footprint in cache. Without kernel dispatch serialization or cross-subsystem synchronization overhead, CPU execution achieves sub-millisecond latency. While batched offline inference (batch size ≥ 32) enables the GPU to amortize dispatch overhead and achieve higher cumulative throughput, for real-time edge streaming where sliding windows arrive sequentially, direct CPU execution provides the optimal low-latency deployment configuration.

5.8. Comparative Transfer Analysis on Strength and Conditioning Exercises (MM-Fit External Benchmark)

To evaluate the cross-dataset generalizability of SkelGym against existing literature, we conducted a rigorous comparative analysis on the multi-sensor MM-Fit dataset [51] aligned with the Strength and Conditioning (S&C) taxonomy of Deyzel et al. [46]. We identified four exact overlapping resistance exercises shared between our benchmark and the external taxonomy: *squat*, *deadlift*, *barbell biceps curl*, and *lateral raise*. We evaluated all baseline architectures and ensembles across 54 held-out external workout recordings ($N = 878$ temporal windows across sessions **w00**, **w05**, **w12**, **w13**, and **w20**) extracted via Google MediaPipe Pose under frozen SkelGym preprocessing and normalization, under both open-set (22 classes) and closed-set (4 classes) settings without any fine-tuning or domain adaptation.

Bayesian Conditional Re-normalization for Closed-Set Evaluation.. Let $\mathcal{C}_{\text{S\&C}} = \{\text{squat, deadlift, barbell biceps curl, lateral raise}\} \subset \mathcal{Y}$ denote the 4-class S&C subset ($|\mathcal{C}_{\text{S\&C}}| = 4$). Given the predicted 22-class posterior probability vector $p = [p_1, \dots, p_{22}] \in \Delta^{21}$ produced by model Θ for window w_k , the closed-set conditional probability for class $c \in \mathcal{C}_{\text{S\&C}}$ is formulated via Bayes’ rule under the hypothesis that the observation belongs to $\mathcal{C}_{\text{S\&C}}$:

$$\tilde{p}_c = P(y = c \mid w_k, y \in \mathcal{C}_{\text{S\&C}}) = \frac{P(y = c \mid w_k)}{\sum_{k' \in \mathcal{C}_{\text{S\&C}}} P(y = k' \mid w_k)} = \frac{p_c}{\sum_{k' \in \mathcal{C}_{\text{S\&C}}} p_{k'}}. \quad (35)$$

The closed-set predicted class is subsequently inferred as $\hat{y}_{\text{closed}} = \arg \max_{c \in \mathcal{C}_{\text{S\&C}}} \tilde{p}_c$.

The performance disparity between open-set and closed-set evaluations in Table 14 directly reflects the structure of the hypothesis space. Under open-set inference over all 22 classes, the random chance baseline is $1/22 \approx 4.55\%$. More critically, the 22-class taxonomy contains fine-grained “kinematic twins” outside $\mathcal{C}_{\text{S\&C}}$ that share nearly identical joint trajectories—most notably *Romanian deadlift* vs. *deadlift* (both sagittal hip hinges), *hammer curl* vs. *barbell biceps curl* (identical elbow flexion differing only in radioulnar pronation), or *shoulder press* vs. *lateral raise* (both shoulder abductions). In open-set inference, minor pose estimation noise or domain shifts can cause probability mass to leak to these competing non-S&C classes, registering as false negatives. Under closed-set conditioning (Eq. (35)), these out-of-subset distractor classes are excluded from the candidate hypothesis space, consolidating probability mass strictly among the four target S&C

Table 14: Comparative transfer evaluation across the four overlapping Strength & Conditioning (S&C) exercises inspired by Deyzel et al. [46] (*squat, deadlift, barbell biceps curl, lateral raise*) evaluated on the genuine MM-Fit external benchmark [51] ($N = 54$ recordings, $N = 878$ windows).

Model Architecture	Open-Set (22 Classes)		Closed-Set S&C Subspace (4 Classes)		
	Test Win Acc	Test Vid Acc	Test Win Acc	Test Vid Acc	Test Vid Macro F1
ST-GCN (Rel 3D) [16] (<i>Deyzel baseline</i>)	37.93%	38.89%	67.43%	70.37%	0.6524
LSTM (Mix 117-d)	61.96%	77.78%	87.02%	90.74%	0.8904
BiLSTM (Mix 117-d)	61.16%	68.52%	80.41%	83.33%	0.7775
AAGCN (Bone 3D)	58.88%	62.96%	74.26%	77.78%	0.6667
Transformer (Mix 117-d)	55.47%	61.11%	83.71%	85.19%	0.8092
SkelGym-Lite (Transformer + Bone)	63.55%	66.67%	79.95%	81.48%	0.7433
SkelGym-Full (Transformer + 4-Stream AAGCN)	66.86%	74.07%	95.90%	100.00%	1.0000

movements (chance level: $1/4 = 25\%$), directly replicating the closed-set experimental setup of Deyzel et al.

Table 14 reports the empirical performance comparison across all seven evaluated models on the MM-Fit external benchmark ($N = 54$ recordings, $N = 878$ windows). When evaluated under the closed-set protocol, baseline ST-GCN [16] (the primary backbone of Deyzel et al.) achieves 67.43% window accuracy and 70.37% video consensus accuracy (Video Macro F1: 0.6524). The recurrent baselines demonstrate strong sequence modeling, with LSTM reaching 87.02% window and 90.74% video accuracy (F1: 0.8904), while BiLSTM achieves 80.41% window and 83.33% video accuracy (F1: 0.7775). Transformer Mix achieves 83.71% window and 85.19% video accuracy (F1: 0.8092), and AAGCN Bone reaches 74.26% window and 77.78% video accuracy (F1: 0.6667). Finally, the proposed multi-stream ensembles establish superior performance gains: SkelGym-Lite reaches 79.95% window and 81.48% video accuracy (F1: 0.7433), while SkelGym-Full achieves an outstanding **95.90% closed-set window accuracy, 100.00% video consensus accuracy, and 1.0000 Macro F1** (correctly classifying all 54 external recordings across the four classes). Under the unconstrained open-set protocol across all 22 classes, SkelGym-Full maintains 66.86% window accuracy and 74.07% video accuracy, demonstrating strong cross-dataset domain robustness without any fine-tuning.

Mitigating the Squat vs. Deadlift Ambiguity (The Deyzel Dilemma). A central limitation reported by Deyzel et al. [46] on the Stellenbosch University Exercise Motion Dataset (SU-EMD) was the intense mutual confusion between back squats and deadlifts: because both compound lower-body exercises involve simultaneous hip and knee flexion-extension in the sagittal plane,

ST-GCN trained on Euclidean Cartesian coordinates exhibited severe diagnostic failure, frequently misclassifying deadlifts as squats. While SU-EMD was constrained to 5 exercises recorded under static laboratory sensor angles, SkelGym scales the taxonomy to 22 fine-grained movements collected across unconstrained, multi-camera gym environments. In our rigorous benchmark on 305 held-out windows across 25 test videos ($N = 15$ squat, $N = 10$ deadlift), baseline ST-GCN fails to disentangle the two kinematics, achieving 0.0% recall on squat and merely 40.0% video recall on deadlift (53.7% window recall), closely mirroring the diagnostic collapse to $\approx 30.0\%$ recall reported on SU-EMD by Deyzel et al. In contrast, by incorporating the 117-dimensional compound feature with hip-to-knee elevation angles $\theta_{\text{hip, knee}}$ and directional bone vectors, Transformer Mix elevates deadlift video recall to 80.0% (squat: 93.3%, with 80.7% window recall), and SkelGym-Full achieves 80.0% deadlift and 86.7% squat video consensus recall under open-set inference (and up to 90.0% deadlift video recall under closed-set consensus), substantially mitigating the mutual confusion without sacrificing specificity.

One-Shot Metric Prototype Classification Simulation Protocol. Deyzel et al. [46] evaluated a 1-shot learning scenario using single-exemplar support matching via cosine similarity, reporting an accuracy of 87.4% on SU-EMD. To benchmark metric generalizability under an identical one-shot evaluation protocol without task-specific fine-tuning or parameter adaptation, we evaluated our pre-trained backbones on the 54 held-out S&C test videos across 100 stochastic trials (random seed 42). In each trial, $K = 1$ random exemplar video was sampled per class as the support set (yielding 4 support videos total across the 4 S&C classes: squat, deadlift, barbell biceps curl, and lateral raise), while the remaining 50 test videos served as query samples. Query classification was executed via nearest-neighbor prototype matching using cosine similarity over L_2 -normalized temporal mean-aggregated feature embeddings. Under this zero-adaptation 1-shot classification simulation, Transformer Mix achieves a mean video accuracy of **90.36% $\pm$ 7.35%** (95% CI: [68.95%, 98.00%], with individual trial peaks reaching **98.00%**), SkelGym-Full reaches **86.18% $\pm$ 9.50%** (95% CI: [60.95%, 98.00%]), and AAGCN Bone reaches **68.82% $\pm$ 11.14%** (95% CI: [48.95%, 89.05%]). These results demonstrate that the proposed 117-d compound representation provides a well-clustered metric space that enables reliable nearest-neighbor recognition even under extreme single-sample constraints, matching and outperforming literature baselines.

Table 15: Taxonomy of failure mechanisms and error mass distribution for the proposed SkelGym-Full ensemble on held-out test partitions ($N = 819$ window errors, $N = 52$ video errors).

Category & Failure Mode	Biomechanical and Optical Etiology	Confused Exercise Classes	Window Errors	Video Errors
Cat 1: Forearm Rotation Ambiguity	Wrist keypoint centroid collapses radioulnar pronation vs. supination rotation	Hammer Curl $\leftrightarrow$ Barbell Biceps Curl	93 (11.36%)	10 (19.23%)
Cat 2: Kinematic Form Overlap	Sagittal hip/knee angle overlap from non-standard athlete lifting execution	Deadlift $\leftrightarrow$ Romanian Deadlift	81 (9.89%)	4 (7.69%)
Cat 3: Perspective Foreshortening	Oblique camera angle compresses vertical torso bench inclination angle	Bench Press $\leftrightarrow$ Incline $\leftrightarrow$ Decline	52 (6.35%)	9 (17.31%)
Cat 4: Closed vs. Open Chain	Identical glenohumeral adduction trajectory in pulling kinetic chains	Lat Pulldown $\leftrightarrow$ Pull-up $\leftrightarrow$ T-Bar Row	82 (10.01%)	9 (17.31%)
Structured Biomechanical Errors	Cumulative impact of top-4 kinematic mechanisms:		308 (37.61%)	32 (61.54%)
Residual Dispersed Errors	Minor uncoordinated errors across remaining 15 classes		511 (62.39%)	20 (38.46%)

6. Biomechanical Kinematics and Error Analysis

A primary objective of this work is to bridge computer vision modeling with clinical biomechanics. Below, we conduct an in-depth anatomical analysis of the model’s predictions and confusion patterns (Fig. 5):

6.1. Taxonomy of Biomechanical Failure Mechanisms

To systematically categorize the error profile of the proposed SkelGym-Full ensemble, Table 15 classifies all misclassifications observed across the held-out test partition ($N = 819$ window errors, $N = 52$ video errors) into four structured physiological mechanisms alongside residual dispersed errors. Remarkably, these four structured biomechanical mechanisms account for **61.54% of all video consensus errors** (32 of 52 misclassified videos) and **37.61% of window-level errors** (308 of 819 windows). This quantitative concentration demonstrates that model confusion is not driven by random stochastic noise, but rather stems from well-defined biomechanical and optical ambiguities inherent to monocular pose estimation in resistance training.

6.2. Upper-Limb Kinematic Ambiguity: Biceps Curl vs. Hammer Curl

Inspection of Fig. 5 indicates that the most prominent confusion occurs between *hammer curl* and *barbell biceps curl* (35.4% recall for hammer curl, with 41.2% of its windows misclassified as biceps curl).

Biomechanically, both exercises execute identical sagittal elbow flexion driven by the elbow joint angle $\theta_{\text{elbow}} = \angle(\text{shoulder}, \text{elbow}, \text{wrist})$ moving

through $30^\circ \rightarrow 145^\circ$. The sole physiological differentiator resides in the forearm’s rotational orientation across the radioulnar joint [44, 45]. In the barbell biceps curl, the forearm is held in full supination (palms oriented anteriorly and superiorly), which maximally tensions the distal tendon insertion across the radial tuberosity and optimizes mechanical leverage for both the long and short heads of the *biceps brachii*. In contrast, the hammer curl maintains a neutral semi-pronated posture (thumbs directed superiorly), shifting primary torque production to the deeper *brachialis* and the superficial *brachioradialis* muscles along the lateral forearm [44, 45]. Because monocular pose estimators (such as MediaPipe) represent the wrist as a single Euclidean point centroid (x, y, z) without recovering full hand rotational mesh orientation or palm surface normals, the 3D trajectory of the wrist joint center in Cartesian space remains virtually identical between supinated and neutral curls. Resolving this subtle anatomical boundary will require integrating fine-grained hand pose keypoints or wrist rotational orientation sensors.

6.3. Posterior Chain Kinematics: Conventional Deadlift vs. Romanian Deadlift

Another noticeable confusion occurs between *deadlift* and *Romanian deadlift* (RDL). The conventional deadlift represents a ground-based compound movement characterized by substantial knee flexion ($\theta_{\text{knee}} \approx 70^\circ - 90^\circ$ at lift initiation) that recruits the quadriceps femoris to generate initial vertical propulsion, transitioning into hip extension as the bar passes the patellae [40]. In contrast, the Romanian deadlift functions as a pure posterior-chain hip hinge movement initiating from an upright standing lockout [41, 39]. Throughout the descent, the knees remain relatively static with slight soft flexion ($\theta_{\text{knee}} \approx 15^\circ - 25^\circ$), with kinematics dominated by maximal anterior pelvic tilt and hip flexion that places intense eccentric loading onto the hamstrings and gluteus maximus. When athletes execute deadlifts with sub-optimal form (e.g., stiff-legged deadlifting with inadequate knee bend) or perform RDLs with excessive knee travel, their sagittal kinematic trajectories overlap significantly. Our 117-d mix feature captures the hip-to-knee elevation angle ($\theta_{\text{hip, knee}}$), enabling the ensemble to separate them correctly in 80% of video consensus cases (Table 12).

6.4. Pressing Plane Angle Analysis: Flat vs. Incline vs. Decline Bench Press

The bench press family targets the *pectoralis major* muscle, with the three variants differing primarily in the inclination angle of the torso plane relative

to the gravity vector [42, 43]. During the flat bench press ($\alpha \approx 0^\circ$), the horizontal pressing trajectory predominantly engages the sternocostal head of the pectoralis major via horizontal glenohumeral adduction. Elevating the bench inclination ($\alpha \approx +30^\circ$ to $+45^\circ$) shifts prime mechanical recruitment superiorly toward the clavicular head of the pectoralis major and anterior deltoids, causing the humerus to elevate along an oblique superior plane relative to the sternum [43]. Conversely, the decline configuration ($\alpha \approx -15^\circ$) isolates the inferior sternocostal fibers with reduced anterior deltoid contribution [42]. Because relative 3D displacement vectors $\tilde{p}$ are referenced to the pelvic hip midpoint, the elevation angles $\theta_{\text{shoulder, wrist}}$ and $\theta_{\text{shoulder, hip}}$ accurately reflect bench tilt. As shown in Table 12, the ensemble achieves 64.29% video recall on flat bench, 62.50% on decline bench, and 0.6154 F1 on incline bench, with errors largely confined to camera angles where bench tilt is foreshortened.

6.5. Overhead and Back Kinetic Chains: Pull-up vs. Lat Pulldown vs. T-Bar Row

A fourth distinct error mass (Category 4, accounting for 17.31% of video errors and 10.01% of window errors) resides in compound pulling exercises targeting the back musculature (*latissimus dorsi*, *trapezius*, and *rhomboids*), predominantly between *pull-up* (Video Recall: 50.00%), *lat pulldown* (Video Recall: 100.00%), and *t-bar row* (Video Recall: 70.00%). Biomechanically, the pull-up represents a closed kinetic chain (CKC) exercise where the athlete’s hands remain fixed to a stationary overhead bar while the torso is propelled vertically upward via glenohumeral adduction and scapular depression. In contrast, the lat pulldown constitutes an open kinetic chain (OKC) exercise where the athlete sits stationary with locked thighs while pulling a resistance bar downward toward the clavicles. While their mechanical boundary conditions differ, both movements execute identical relative angular trajectories across the shoulders and elbows: the humerus adducts in the frontal plane from $170^\circ \rightarrow 30^\circ$ relative to the torso, while the elbow flexes from $180^\circ \rightarrow 70^\circ$. Because relative 3D coordinate centering references all joint positions to the pelvic hip midpoint, the absolute translation of the body relative to the external room environment is factored out. Consequently, during execution phases where the athlete’s torso remains vertically upright, the relative spatial coordinates and pairwise angles between pull-ups and lat pulldowns become nearly identical, leading to occasional misclassifications under oblique camera perspectives.

6.6. Validation-Grounded Decision Protocol

We emphasize that all model selection choices—including adopting the 117-d mix feature, retaining SkelGym-Aug, and solving SLSQP voting weights—were established strictly on the validation set, where the SkelGym-Full ensemble reached 80.10% window accuracy and 83.09% video accuracy. The subsequent $78.83\% \pm 0.66\%$ test consensus result (78.97% on the baseline run) represents a reliable held-out confirmation of generalization to unseen subjects and recording environments. Finally, out of 236 held-out test videos, exactly 233 contained valid action segments exceeding 32 frames after boundary trimming, explaining the test support count.

7. Discussion, Limitations, and Future Work

Trade-offs Between Kinematic Skeleton Fusion and Dense Video Vision Transformers.. While SkelGym deliberately centers on lightweight, coordinate-based 3D skeleton sequences, an alternative foundational paradigm in computer vision relies on dense spatio-temporal video architectures, including 3D CNNs (e.g., SlowFast [3]), Video Vision Transformers (e.g., Video Swin Transformer / Swin3D [6]), and volumetric heatmap convolutional networks (e.g., PoseConv3D [56]). Dense video transformers possess the distinct theoretical advantage of directly ingesting raw appearance pixels, allowing them to exploit environmental context, gym machine geometries, barbell grip orientations, and subtle muscular contractions without relying on intermediate landmark extraction pipelines. This appearance awareness is particularly beneficial for resolving fine-grained ambiguities such as the pronation vs. supination distinction between hammer curls and barbell curls.

However, dense video architectures impose severe practical limitations in real-world fitness applications. First, their computational complexity is substantial: Video Swin models frequently require tens to hundreds of GFLOPs per snippet, incurring latencies of several hundred milliseconds per prediction window and rendering them completely unsuited for real-time inference on edge smartphones or micro-controllers. In contrast, SkelGym executes in 0.42–3.96 ms on host CPU (0.50–4.65 ms on CUDA) per window. Second, RGB video models exhibit severe vulnerability to environmental background and athletic apparel memorization, leading to dramatic generalization collapse when evaluated in unseen home gym settings. Third, cloud-based video analytics require continuously transmitting unconstrained, high-resolution RGB living-room video streams across public networks, introduc-

ing severe security vulnerabilities and privacy intrusion concerns for consumer smart-camera devices. In contrast, while monocular pose estimation (such as MediaPipe) still operates on optical camera frames at the physical sensor level, the ultra-low latency of compact skeleton classifiers enables the entire perception pipeline to execute locally on edge hardware. Under this edge-native paradigm, raw RGB frames are processed ephemerally in volatile RAM and immediately discarded frame-by-frame without persistent disk caching or remote network streaming. Only low-dimensional coordinate trajectories (x, y, z) are retained, enabling a privacy-aware workflow that prevents continuous cloud video streaming. We emphasize that while 3D skeletal coordinates eliminate optical identity markers (such as facial features, body textures, clothing, and domestic background surroundings), they are not completely anonymous, as coarse anthropometric proportions (e.g., relative limb lengths) and movement signatures remain partially discernible in continuous trajectories; nonetheless, local coordinate extraction provides a substantial privacy barrier compared to unconstrained RGB streaming. Consequently, we identify hybrid hierarchical architectures—wherein lightweight skeleton models handle primary real-time classification, triggering localized, low-frequency 3D video attention only when kinematic ambiguity surpasses uncertainty thresholds—as a promising direction for future research.

Edge Feasibility vs. Heavy Modern Skeleton Transformers (CTR-GCN / SkateFormer). Recent advances in general-purpose action recognition have produced capable spatial-temporal architectures, including Channel-Wise Topology Refinement GCN (CTR-GCN) [20] and skeletal transformers like SkateFormer [21]. While achieving top-tier performance on large benchmarks such as NTU RGB+D [32], they are typically tailored for full-body 25-joint topologies with dense hand-finger and multi-segment spine keypoints. More critically, modern skeleton transformers incur large parameter footprints (often exceeding several million parameters) and heavy computational requirements, frequently requiring pre-training on massive external action databases. In contrast, automated resistance training feedback demands ultra-low latency on edge consumer hardware (e.g., smart mirrors, mobile devices, and gym-floor tablets) operating under strict thermal and battery constraints. When deployed in high-speed streaming scenarios (30 FPS, 33.3 ms real-time budget), heavy transformer models introduce significant latency and compute overhead. SkelGym addresses this operational bottleneck through custom-engineered lightweight architectures ($\approx 350\text{K}$ param-

ters per backbone stream, 1.91M parameters total for the full 5-stream ensemble, and only 778K parameters for SkelGym-Lite) trained entirely from scratch with zero pretraining overhead. With single-window classifier inference executing in merely 0.42–3.96 ms on host CPU (0.50–4.65 ms on CUDA, where individual backbones require 0.42–1.03 ms and the full 5-stream ensemble executes in 3.96 ms), SkelGym achieves an optimal Pareto balance between high fine-grained exercise recognition accuracy and real-time edge execution feasibility.

Limitations of Monocular Pose Estimation and Failure Modes.. While monocular pose extraction via MediaPipe Pose provides high throughput on edge processors, it incurs physical and optical failure modes that bound skeleton-based recognition accuracy. First, resistance training environments frequently introduce severe physical occlusions: barbell weight plates (e.g., standard 450 mm Olympic plates) frequently obstruct wrist, elbow, and torso tracking during deadlifts, rows, and bench presses, while spotters standing adjacent to equipment introduce visual occlusion. Second, camera field-of-view boundaries cause transient out-of-frame limb clipping, such as feet moving below frame during Romanian deadlifts or hands ascending above frame during pull-ups and overhead presses. Third, monocular 2D-to-3D landmark lifting estimates metric depth (z) relative to the hip midpoint using geometric heuristics and anthropometric priors. Consequently, movements aligned along the camera’s optical axis exhibit higher longitudinal depth jitter than lateral sagittal plane perspectives. Future iterations could incorporate multi-view spatial consensus or temporal smoothing filters (e.g., learned kinematic Kalman filters) to improve landmark stability.

Dataset Class Imbalance and Long-Tail Kinetics.. A second practical limitation concerns the natural class distribution within crowdsourced fitness video datasets. Movement availability reflects gym training popularity: high-frequency compound and aesthetic movements (such as barbell biceps curls and bench presses, each with 84 videos) possess extensive diversity, whereas specialized or bodyweight stabilization exercises (such as plank with 9 videos) reside in the long tail. Although we mitigated class imbalance during optimization using Macro F1 monitoring, validation-calibrated loss weighting, and label smoothing ($\alpha = 0.05$), rare classes exhibit wider confidence intervals. Expanding the public corpus through standardized multi-angle video collection across underrepresented exercises remains an essential objective for subsequent dataset releases.

Conclusion. In this work, we presented **SkelGym**, a lightweight, computationally efficient, and biomechanically grounded intelligent exercise recognition system for 22-class gym exercise classification. By coupling the proposed 117-dimensional Biomechanical Compound Representation with the empirically selected SkelGym-Aug configuration (unifying bilateral reflection, gravitational yaw rotation, proportional scaling, and sensor jitter) and integrating self-attention sequence modeling with 4-stream adaptive graph convolutions via validation-calibrated soft voting, SkelGym attains **69.73% $\pm$ 1.10% window-level accuracy** and **78.83% $\pm$ 0.66% video-level consensus accuracy** (Video Macro F1: **0.7807 $\pm$ 0.0126**, Window Macro F1: 0.6881 ± 0.0072) across three independent random seeds under fixed partitions across 22 resistance exercises, with selected model comparisons verified via paired statistical tests (McNemar $p < 10^{-14}$, Wilcoxon $p < 0.005$). With an ultra-low single-window classifier inference latency of **0.42–3.96 ms on CPU** (0.50–4.65 ms on CUDA, 1.24–5.86 ms on MPS; where individual backbones require 0.42–1.03 ms and the full 5-stream ensemble executes in 3.96 ms) per 32-frame window, SkelGym enables real-time deployment on resource-constrained edge devices including smart mirrors, mobile phones, and embedded gym displays. By operating exclusively on privacy-aware 3D skeletal coordinates rather than raw RGB video streams, SkelGym eliminates the need for continuous cloud video transmission, making it suitable for privacy-sensitive home and commercial gym environments. Our comprehensive evaluation on in-the-wild multi-camera videos—rather than laboratory-controlled settings alone—demonstrates that the proposed pipeline offers a robust, practical foundation for next-generation intelligent virtual personal trainer systems.

Data, Reproducibility, and Code Availability

To facilitate rigorous scientific reproducibility and edge deployment, all project software, extracted kinematic representations, and pre-trained models are publicly hosted on Hugging Face Hub at <https://huggingface.co/Cuong2004/gym-exercise-classification>. The repository includes PyTorch source code for all sequence and spatial-temporal graph backbones (Transformer Mix, ST-GCN, and Four-Stream AAGCN), training and evaluation scripts, and trained model checkpoints.

The complete skeletal dataset is released in JSON, NPY, and HDF5 tensor formats, accompanied by exact temporal action segment boundaries

(.txt) and consolidated metadata records (Final_dataset_metadata.csv) across Hugging Face and Kaggle. Distributing pre-extracted 3D landmark coordinates directly eliminates reliance on third-party video URLs that suffer from link-rot, while ensuring deterministic reproducibility without demanding intensive GPU pose estimation re-runs. In strict compliance with online platform Terms of Service and subject privacy considerations, raw third-party internet video recordings are withheld from redistribution. However, the 128 author-recorded gymnasium video clips (collected with informed consent under institutional ethical oversight) and all 1,024 extracted coordinate sequences are freely accessible for academic and non-commercial research purposes.

Exact Reproducibility Protocol. To enable deterministic replication of every result reported in this paper, the repository provides self-contained evaluation scripts that operate directly on pre-trained checkpoints without requiring model retraining:

```
git clone \
  https://huggingface.co/Cuong2004/gym-exercise-classification
pip install -r requirements.txt
# Evaluation suite across Tables 1-7:
python scripts/evaluate_local_ensemble.py
# External S&C benchmark (Table 8):
python scripts/evaluate_external_benchmark.py
# Statistical significance tests (Table 11):
python scripts/compute_statistical_tests.py
# Latency & FLOPs benchmarking (Table 12):
python scripts/benchmark_hardware_latency.py
```

All experiment configuration files (YAML configs under `configs/experiments/`), dataset metadata (Final_dataset_metadata.csv), partition manifests, and pre-trained PyTorch checkpoints for every backbone variant (Transformer Mix, ST-GCN, Four-Stream AAGCN Joint/Bone/Joint-Motion/Bone-Motion) are publicly hosted on Hugging Face Model Hub. This ensures 100% reproducibility of all reported tables without retraining from scratch.

CRediT Author Contributions

Xuan-Cuong Nguyen: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Nhat-Quang Truong:** Data curation, Resources, Software, Validation, Writing – review & editing. **Quoc-Trinh Vo:** Supervision, Formal analysis, Validation, Writing – review & editing.

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Appendix A. Anatomical Landmark Mapping and Physiological Significance

Table A.16 outlines the 13 anatomical keypoints selected from Google MediaPipe Pose Heavy [8, 9]. These joints constitute the core kinematic chain required for analyzing human resistance exercises. Facial landmarks (1–10) and foot extremity points (29–32) are intentionally discarded to eliminate high-frequency tracking noise.

Table A.16: Selected 13 anatomical joints from MediaPipe Pose and their kinematic roles.

Joint ID (v)	MediaPipe Index	Anatomical Description	Kinematic Significance in Resistance Training
0	0	Nose	Cranial orientation tracking
1	11	Left Shoulder	Glenohumeral joint; upper-body pressing/pulling rotation center
2	12	Right Shoulder	Bilateral counterpart to left shoulder
3	13	Left Elbow	Humeroradial joint; elbow flexion/extension in curls/presses
4	14	Right Elbow	Bilateral counterpart to left elbow
5	15	Left Wrist	Radiocarpal joint; barbell/dumbbell contact point trajectory
6	16	Right Wrist	Bilateral counterpart to left wrist
7	23	Left Hip	Acetabulofemoral joint; pelvic hinge pivot; midpoint with Right Hip defines coordinate origin anchor
8	24	Right Hip	Bilateral counterpart to left hip; midpoint with Left Hip defines coordinate origin anchor
9	25	Left Knee	Tibiofemoral joint; knee flexion/extension in squats/lunges
10	26	Right Knee	Bilateral counterpart to left knee
11	27	Left Ankle	Talocrural joint; ground contact anchor; ankle dorsiflexion
12	28	Right Ankle	Bilateral counterpart to left ankle

Appendix B. Open Science Model Hub and Checkpoint Loading

All pre-trained PyTorch [30] checkpoints, scripts, and evaluation pipelines are openly accessible on Hugging Face at <https://huggingface.co/Cuong2004/gym-exercise-classification>. Checkpoints can be downloaded in Python:

```
import torch
from huggingface_hub import hf_hub_download
```

```
# Download best Transformer Mix checkpoint
ckpt_path = hf_hub_download(
    repo_id="Cuong2004/gym-exercise-classification",
    filename="best_transformer_mix_aug.pt"
)
model = torch.load(ckpt_path, map_location="cpu")
print("Model loaded successfully!")
```